

Bloch

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Started 15th June 2026

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# Chapter 1

## Construction of the Real Numbers

**Definition 1.1.** Let  $S$  be a set. A **binary operation** on  $S$  is a function  $S \times S \rightarrow S$ . A **unary operation** on  $S$  is a function  $S \rightarrow S$ .

Let  $S$  be a set and let  $*$  :  $S \times S \rightarrow S$  be a binary operation. If  $x, y \in S$ , the correct way to write the result of the operation to the pair  $\langle x, y \rangle$  would be  $*(\langle x, y \rangle)$ . However since this is a bit cumbersome one may write  $x * y$  instead. Similarly if  $\neg$  :  $S \rightarrow S$  is a unary operation and  $x \in S$  one may choose to write  $\neg x$  over  $\neg(x)$ .

### 1.1 Axioms for the Natural numbers

#### 1.1.1 Peano Postulates

*Axiom: Peano Postulates*

There exists a set  $\mathbb{N}$  with an element  $1 \in \mathbb{N}$  and a function  $s : \mathbb{N} \rightarrow \mathbb{N}$  that satisfy the following three properties.

- There is no  $n \in \mathbb{N}$  such that  $s(n) = 1$ .
- The function  $s$  is injective.
- Let  $G \subseteq \mathbb{N}$  be a set. Suppose that  $1 \in G$ , and that if  $g \in G$  then  $s(g) \in G$ . Then  $G = \mathbb{N}$ .

It will be shown that the set  $\mathbb{N}$  is unique. We can therefore give it a name.

**Definition 1.2.** The set of **natural numbers** denoted  $\mathbb{N}$ , is the set the existence of which is given in the Peano Postulates.

### 1.1.2 Consequences of Peano Postulates

**Lemma 1.3.** *Let  $a \in \mathbb{N}$ . Suppose that  $a \neq 1$ . Then there is a unique  $b \in \mathbb{N}$  such that  $a = s(b)$ . (where existence of  $s$  and  $\mathbb{N}$  are outlined by the peano axioms)*

*Proof.* First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that  $G = \mathbb{N}$ . To do this we need to show  $1 \in G$ ,  $G \subseteq \mathbb{N}$  and  $\forall g \in G(s(g) \in G)$ . The first two are straightforward. For the third suppose  $g \in G$ ; we have two cases. First suppose  $g \in \{1\}$ ; then  $g = 1 \in \mathbb{N}$ . Since  $s : \mathbb{N} \rightarrow \mathbb{N}$ , we then have  $s(1) \in \mathbb{N}$ . So by definition  $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$  and  $s(g) = s(1) \in G$ . Now suppose  $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ ; then by definition  $g \in \mathbb{N}$  and  $s(g) \in \mathbb{N}$  exists. By definition  $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$  so  $s(g) \in G$ . By the third peano postulate we then can assert that  $G = \mathbb{N}$ . So  $a \in \mathbb{N} = G$ . Since  $a \neq 1$  we must then have  $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$  and by definition,  $\exists b \in \mathbb{N}(s(b) = a)$ .

Now for uniqueness. Let  $m$  and  $n$  be arbitrary elements of  $\mathbb{N}$  such that  $a = s(m)$  and  $a = s(n)$ . Then  $\langle m, a \rangle \in s$  and  $\langle n, a \rangle \in s$ . Since  $s$  is injective we must have  $n = m$ .  $\square$

**Theorem 1.4.** *Let  $H$  be a set, Let  $e \in H$  and let  $k : H \rightarrow H$  be a function. Then there is a unique function  $f : \mathbb{N} \rightarrow H$  such that  $f(1) = e$  and  $f \circ s = k \circ f$ .*

*Proof.* Letting  $H, e$ , and  $k$  be arbitrary such that  $e \in H$  and  $k : H \rightarrow H$ . We will show existence and uniqueness separately. We start with uniqueness.

Let  $g$  and  $h$  be arbitrary such that  $g : \mathbb{N} \rightarrow H$ ,  $g(1) = e$ , and  $g \circ s = k \circ g$  and  $h : \mathbb{N} \rightarrow H$ ,  $h(1) = e$ , and  $h \circ s = k \circ h$ . We will first define  $V$  as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that  $V = \mathbb{N}$ . By peano's postulates, we need to show  $V \subseteq \mathbb{N}$ ,  $1 \in V$ , and  $\forall n \in V(s(n) \in V)$ . The first is straightforward. For the second requirement, we know by peano's postulates that  $1 \in \mathbb{N}$ , so by our assumptions  $g(1)$  and  $h(1)$  exist and  $g(1) = e = h(1)$ . So  $1 \in V$ .

For the third requirement, let  $n$  be arbitrary and suppose that  $n \in V$ . Then by definition  $n \in \mathbb{N}$  and  $g(n) = h(n)$ . Since  $n \in \mathbb{N} = \text{dom}(h \circ s)$ , we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition  $\text{ran}(s) \subseteq \mathbb{N}$ , so  $s(n) \in \mathbb{N}$  and again by definition  $s(n) \in V$ . We can conclude that  $V = \mathbb{N}$ .

Suppose  $\langle x, y \rangle \in g$ , then  $x \in \text{dom} g = \mathbb{N} = V$ . So by definition  $y = g(x) = h(x)$ , and  $\langle x, y \rangle \in h$ . Suppose  $\langle x, y \rangle \in h$ , then  $x \in \text{dom} h = \mathbb{N} = V$ . So by definition  $y = h(x) = g(x)$ , and  $\langle x, y \rangle \in g$ .

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Now for existence. We need to come up with a function from  $\mathbb{N}$  into  $H$  such that  $f(1) = e$  and  $f \circ s = k \circ f$ . We first define  $C$ , where

$$\forall X (X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e \rangle \in X \wedge \forall n \forall y (\langle n, y \rangle \in X \rightarrow \langle s(n), k(y) \rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that  $f = \bigcap C$  is the desired function. To ensure its existence, we need to show that  $C$  is nonempty. We will show that  $\mathbb{N} \times H \in C$ . For the first requirement, clearly  $\mathbb{N} \times H \subseteq \mathbb{N} \times H$ . Next, since  $1 \in \mathbb{N}$  and  $e \in H$  then  $\langle 1, e \rangle \in \mathbb{N} \times H$ . Finally, let  $n$  and  $y$  be arbitrary such that  $\langle n, y \rangle \in \mathbb{N} \times H$ . Since  $s : \mathbb{N} \rightarrow \mathbb{N}$  and  $k : H \rightarrow H$ , we must then have  $s(n) \in \mathbb{N}$  and  $k(y) \in H$ . So  $\langle s(n), k(y) \rangle \in \mathbb{N} \times H$ .  $f = \bigcap C$  is therefore a valid candidate.

Before we prove the functionality of  $f$ , we first state a few facts about  $f$ . First we note that  $f \in C$ . Let arbitrary  $x \in \bigcap C$ . Since  $C$  is nonempty there exists  $X \in C$ , and by definition  $x \in X \subseteq \mathbb{N} \times H$ . So  $\bigcap C \subseteq \mathbb{N} \times H$ . Next let arbitrary  $X \in C$ . By definition  $\langle 1, e \rangle \in X$ , so  $\langle 1, e \rangle \in \bigcap C$ . Finally let  $n$  and  $y$  be arbitrary and suppose  $\langle n, y \rangle \in \bigcap C$ . We need to show that  $\langle s(n), k(y) \rangle \in \bigcap C$ . Let arbitrary  $X \in C$ . We have  $\langle n, y \rangle \in X$ . Since  $X \in C$ , by definition we must then have  $\langle s(n), k(y) \rangle \in X$ . We are done.

Next we note that  $\forall X \in C (f \subseteq X)$ . This is straightforward to prove. Finally we show

$$\begin{aligned} \forall n \forall y ([\langle n, y \rangle \in f \wedge \langle n, y \rangle \neq \langle 1, e \rangle] \\ \rightarrow \exists m \exists u (\langle m, u \rangle \in f \wedge \langle n, y \rangle = \langle s(m), k(u) \rangle)) \end{aligned}$$

Let  $n$  and  $y$  be arbitrary such that  $\langle n, y \rangle \in f$  and  $\langle n, y \rangle \neq \langle 1, e \rangle$ . Suppose, for contradiction, that the statement after the implication is false. Then  $\forall m \forall u (\langle m, u \rangle \in f \rightarrow \langle n, y \rangle \neq \langle s(m), k(u) \rangle)$ . The set  $f - \{\langle n, y \rangle\}$  exists by the subset axiom; we will show that this set is a member of  $C$ . We know that  $f \in C$ , so  $f \subseteq \mathbb{N} \times H$ , and it is straightforward to show that  $(f - \{\langle n, y \rangle\}) \subseteq \mathbb{N} \times H$ . Since  $\langle 1, e \rangle \neq \langle n, y \rangle$  and  $\langle 1, e \rangle \in f$ , we also have  $\langle 1, e \rangle \in f - \{\langle n, y \rangle\}$ . Finally, let arbitrary  $\langle a, b \rangle \in f - \{\langle n, y \rangle\}$ . Then  $\langle a, b \rangle \in f \in C$  and  $\langle s(a), k(b) \rangle \in f$ . Since  $\langle a, b \rangle \in f$ , we must have, by our assumption,  $\langle n, y \rangle \neq \langle s(a), k(b) \rangle$  and  $\langle s(a), k(b) \rangle \in f - \{\langle n, y \rangle\}$ . So  $f - \{\langle n, y \rangle\} \in C$ . It is straightforward to show that  $f \not\subseteq f - \{\langle n, y \rangle\}$ , contradicting the fact that  $\forall X \in C (f \subseteq X)$ .

We now prove functionality. We will do this by defining  $G$ :

$$\forall z (z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H (\langle z, x \rangle \in f))$$

(which exists by the subset axiom) and using the peano postulates to show that  $G = \mathbb{N}$ .

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We defined  $G$ :

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x \rangle \in f))$$

To show that  $G = \mathbb{N}$ , we need to show that  $G \subseteq \mathbb{N}$ ,  $1 \in G$ , and  $\forall n \in G(s(n) \in G)$ . The first requirement is straightforward. For the second, we already know by definition that  $1 \in \mathbb{N}$ . Since  $f \in C$  we know by definition that  $\langle 1, e \rangle \in f$ , where  $e \in H$ . Let arbitrary  $x' \in H$  such that  $\langle 1, x' \rangle \in f$ . Suppose for contradiction that  $e \neq x'$ . Then we have  $\langle 1, e \rangle \neq \langle 1, x' \rangle$ ; so by the third statement proven earlier there exists  $m, u$  such that  $\langle m, u \rangle \in f$  and  $\langle 1, x' \rangle = \langle s(m), k(u) \rangle$ . This means that  $1 = s(m)$ , where since  $f \subseteq \mathbb{N} \times H$ ,  $m \in \mathbb{N}$ . This contradicts the peano postulates, which assert that  $\neg \exists n \in \mathbb{N}(s(n) = 1)$ .

For the third and final requirement, let  $n$  be an arbitrary element of  $G$ . Then by definition  $n \in \mathbb{N}$  and there exists a unique  $x$  such that  $\langle n, x \rangle \in f$ . By definition we know  $s(n) \in \mathbb{N}$ , and since  $f \in C$ , we have  $\langle s(n), k(x) \rangle \in f$ , this verifies existence. Now let  $x'$  be arbitrary such that  $\langle s(n), x' \rangle \in f$ . By peano's postulates we know that  $s(n) \neq 1$ , so  $\langle s(n), x' \rangle \neq \langle 1, e \rangle$ . By the third statement proven earlier there exists  $m, u$  such that  $\langle m, u \rangle \in f$  and  $\langle s(n), x' \rangle = \langle s(m), k(u) \rangle$ . So  $s(n) = s(m)$  and  $x' = k(u)$ . Since  $s$  is injective we must have  $m = n$ . So  $\langle n, u \rangle \in f$ . By uniqueness of  $x$  we then have  $u = x$ . So  $x' = k(u) = k(x)$ , proving uniqueness.  $s(n) \in G$ .

We have shown that  $G = \mathbb{N}$ . It follows from this that  $f : \mathbb{N} \rightarrow H$ . For functionality, let  $x \in \text{dom}f$ . Then there exists  $y$  such that  $\langle x, y \rangle \in f$ . Since  $f \subseteq \mathbb{N} \times H$  we have  $x \in \mathbb{N} = G$ , and by definition  $y$  is unique. Next to show  $\text{dom}f = \mathbb{N}$ , first let arbitrary  $x \in \text{dom}f$ , similarly it immediately follows that  $x \in \mathbb{N}$ . Now let  $x \in \mathbb{N}$ . Then  $x \in G$  and by definition there exists  $y \in H$  such that  $\langle x, y \rangle \in f$ , meaning  $x \in \text{dom}f$ . Finally to show  $\text{ran}f \subseteq H$ , let  $y \in \text{ran}f$ . Then there exists  $x$  such that  $\langle x, y \rangle \in f \subseteq \mathbb{N} \times H$ , so  $y \in H$ .

The fact that  $\langle 1, e \rangle \in \bigcap C = f$  was already shown earlier. Finally we show  $f \circ s = k \circ f$ . Let  $\langle x, y \rangle \in f \circ s$ . Then  $y = f \circ s(x) = f(s(x))$ . Since  $s(x) \neq 1$ ,  $\langle s(x), y \rangle \neq \langle 1, e \rangle$  and as proven earlier there then exists  $m, u$  such that  $\langle m, u \rangle \in f$  and  $\langle s(x), y \rangle = \langle s(m), k(u) \rangle$ . So  $s(x) = s(m)$  and  $y = k(u)$ . Since  $s$  is injective  $x = m$ . We then have  $\langle x, u \rangle \in f$  and  $y = k(u) = k(f(x)) = k \circ f(x)$ . so  $\langle x, y \rangle \in k \circ f$ . Next, suppose  $\langle x, y \rangle \in k \circ f$ . By definition there exists  $z$  such that  $\langle x, z \rangle \in f$  and  $\langle z, y \rangle \in k$ . Since  $f \in C$  we have  $\langle s(x), k(z) \rangle \in f$ . We then have  $y = k(z) = f(s(x)) = f \circ s(x)$  and  $\langle x, y \rangle \in f \circ s$ . We are done.  $\square$

### 1.1.3 Addition on $\mathbb{N}$

**Theorem 1.5.** *There is a unique binary operation  $+$  :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  that satisfies the following two properties for all  $n, m \in \mathbb{N}$ :*

- $n + 1 = s(n)$
- $n + s(m) = s(n + m)$

*Proof.* We start with uniqueness. Let  $+$  and  $\oplus$  be arbitrary such that  $+, \oplus : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ ,  $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (+( \langle n, 1 \rangle ) = s(n) \wedge +( \langle n, s(m) \rangle ) = s(+( \langle n, m \rangle )))$ , and  $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\oplus( \langle n, 1 \rangle ) = s(n) \wedge \oplus( \langle n, s(m) \rangle ) = s(\oplus( \langle n, m \rangle )))$ . We need to show  $+$  =  $\oplus$ . We define  $G$  as

$$\forall x (x \in G \leftrightarrow x \in \mathbb{N} \wedge \forall n \in \mathbb{N} (+( \langle n, x \rangle ) = \oplus( \langle n, x \rangle )))$$

(guaranteed by the subset axiom.) We will show that  $G = \mathbb{N}$ . To do this we need to show  $G \subseteq \mathbb{N}$ ,  $1 \in G$ , and  $\forall g \in G (s(g) \in G)$ . The first requirement is straightforward. For the second requirement, we already know  $1 \in \mathbb{N}$ , and for arbitrary  $n \in \mathbb{N}$ ,  $+( \langle n, 1 \rangle ) = s(n) = \oplus( \langle n, 1 \rangle )$ .

For the third requirement, let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $\forall n \in \mathbb{N} (+( \langle n, g \rangle ) = \oplus( \langle n, g \rangle ))$ . Since  $s : \mathbb{N} \rightarrow \mathbb{N}$  we know that  $s(g) \in \mathbb{N}$  exists. Let  $n$  be arbitrary. By definition  $+( \langle n, s(g) \rangle ) = s(+( \langle n, g \rangle ))$  and  $\oplus( \langle n, s(g) \rangle ) = s(\oplus( \langle n, g \rangle ))$ . By assumption we can assert that  $+( \langle n, g \rangle ) = \oplus( \langle n, g \rangle )$ , so  $+( \langle n, s(g) \rangle ) = s(+( \langle n, g \rangle )) = s(\oplus( \langle n, g \rangle )) = \oplus( \langle n, s(g) \rangle )$ . We conclude that  $s(g) \in G$  and  $G = \mathbb{N}$ .

Let  $\langle \langle x, y \rangle, z \rangle \in +$ . By definition  $y \in \mathbb{N} = G$  and  $x \in \mathbb{N}$ . So  $+( \langle x, y \rangle ) = \oplus( \langle x, y \rangle )$  and  $z = \oplus( \langle x, y \rangle )$ . So  $\langle \langle x, y \rangle, z \rangle \in \oplus$ . Now let  $\langle \langle x, y \rangle, z \rangle \in \oplus$ . A similar argument shows that  $\langle \langle x, y \rangle, z \rangle \in +$ .

Now for existence. We want to show

$$\begin{aligned} \exists + ( + : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N} ( &+( \langle n, 1 \rangle ) = s(n) \\ &\wedge +( \langle n, s(m) \rangle ) = s(+( \langle n, m \rangle ))) ) \end{aligned}$$

Theorem 1.4 allows us to assert

$$\forall k \forall H \forall e ([e \in H \wedge k : H \rightarrow H] \rightarrow \exists ! f (f : \mathbb{N} \rightarrow H \wedge f(1) = e \wedge f \circ s = k \circ f))$$

It immediately follows that

$$\forall p \in \mathbb{N} \exists ! f (f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

We will define  $+$  as the set where

$$\begin{aligned} \forall z (z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} (\exists f (f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c (\langle b, c \rangle \in f \wedge z = \langle \langle a, b \rangle, c \rangle ))) ) \end{aligned}$$

Existence follows from the subset axiom, where it can be shown that  $+$   $\subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ .

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We defined  $+$  as

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} (\exists f(f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \wedge \exists c(\langle b, c \rangle \in f \wedge z = \langle \langle a, b \rangle, c \rangle))))$$

First we prove functionality. Let  $x$  be an arbitrary element of  $\text{dom}(+)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in +$ . Let  $y'$  be arbitrary such that  $\langle x, y' \rangle \in +$ .  $\langle x, y \rangle \in +$  by definition means that there exists  $a, b \in \mathbb{N}$ , and  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that  $f(1) = s(a)$  and  $f \circ s = s \circ f$ , and that there exists  $c$  such that  $\langle b, c \rangle \in f$  and  $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$ .  $\langle x, y' \rangle \in +$  similarly implies the existence of  $a_0, b_0 \in \mathbb{N}$  and  $f_0 : \mathbb{N} \rightarrow \mathbb{N}$  such that  $f_0(1) = s(a_0)$  and  $f_0 \circ s = s \circ f_0$ , and the existence of  $c_0$  such that  $\langle b_0, c_0 \rangle \in f_0$  and  $\langle x, y' \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$ . Then  $\langle a, b \rangle = x = \langle a_0, b_0 \rangle$ , so  $a_0 = a$  and  $b_0 = b$ . It follows that  $s(a_0) = s(a) \in \mathbb{N}$ . This and

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \rightarrow \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

implies that  $f = f_0$ . So we have  $y = c = f(b) = f(b_0) = f_0(b_0) = c_0 = y'$ .

Next we show that  $\text{dom}(+) = \mathbb{N} \times \mathbb{N}$ . First let  $x$  be an arbitrary element of  $\text{dom}(+)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in +$ . By definition there then exists  $a, b \in \mathbb{N}$  and there exists  $c$  such that  $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$ . So  $x = \langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$ .

Now suppose  $x \in \mathbb{N} \times \mathbb{N}$ . Then by definition there exists  $a, b \in \mathbb{N}$  such that  $x = \langle a, b \rangle$ .  $s(a) \in \mathbb{N}$  exists, so by theorem 1.4 there exists  $f$  such that  $f : \mathbb{N} \rightarrow \mathbb{N}$ ,  $f(1) = s(a)$ , and  $f \circ s = s \circ f$ . Since  $b \in \mathbb{N} = \text{dom} f$ ,  $f(b)$  exists and  $\langle b, f(b) \rangle \in f$ . Then by definition,  $\langle \langle a, b \rangle, f(b) \rangle \in +$ , so  $x = \langle a, b \rangle \in \text{dom}(+)$ .

Next we show that  $\text{ran}(+) \subseteq \mathbb{N}$ . Let arbitrary  $y \in \text{ran}(+)$ . Then there exists  $x$  such that  $\langle x, y \rangle \in +$ . By definition there then exists  $a, b \in \mathbb{N}$ ,  $f : \mathbb{N} \rightarrow \mathbb{N}$ , and  $c$  such that  $\langle b, c \rangle \in f$  and  $\langle x, y \rangle = \langle \langle a, b \rangle, c \rangle$ . So we have  $y = c = f(b) \in \mathbb{N}$ .

Now we verify the two properties of addition, that is,

$$\forall m \in \mathbb{N} \forall n \in \mathbb{N} (+(\langle n, 1 \rangle) = s(n) \wedge +(\langle n, s(m) \rangle) = s(+(\langle n, m \rangle)))$$

Let arbitrary  $n, m \in \mathbb{N}$ . We know that  $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ , so  $\langle n, 1 \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$  and there exists  $y$  such that  $\langle \langle n, 1 \rangle, y \rangle \in +$ . By definition there then exists  $a, b \in \mathbb{N}$  and  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that  $f(1) = s(a)$ , and there exists  $c$  such that  $\langle b, c \rangle \in f$  and  $\langle \langle n, 1 \rangle, y \rangle = \langle \langle a, b \rangle, c \rangle$ . Then  $n = a$ ,  $1 = b$ , and  $y = c$ . We then have  $\langle 1, y \rangle \in f$  and  $y = f(1) = s(a) = s(n)$ . We conclude that  $\langle \langle n, 1 \rangle, s(n) \rangle \in +$  and  $+(\langle n, 1 \rangle) = s(n)$ .

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Now for the second property. We know that  $s(m) \in \mathbb{N}$ . So  $\langle n, s(m) \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$  and there exists  $y$  such that  $\langle \langle n, s(m) \rangle, y \rangle \in +$ . By definition there then exists  $a, b \in \mathbb{N}$  such that  $f : \mathbb{N} \rightarrow \mathbb{N}$ ,  $f(1) = s(a)$ , and  $f \circ s = s \circ f$ , and there exists  $c$  such that  $\langle b, c \rangle \in f$  and  $\langle \langle n, s(m) \rangle, y \rangle = \langle \langle a, b \rangle, c \rangle$ . We then have  $n = a$ ,  $s(m) = b$ , and  $y = c = f(b) = f(s(m)) = s(f(m))$ . We know that  $\langle n, m \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ , so there exists  $y_0$  such that  $\langle \langle n, m \rangle, y_0 \rangle \in +$ . Again by definition there exists  $a_0, b_0 \in \mathbb{N}$  and  $f_0 : \mathbb{N} \rightarrow \mathbb{N}$  such that  $f_0(1) = s(a_0)$  and  $f_0 \circ s = s \circ f_0$  and there exists  $c_0$  such that  $\langle b_0, c_0 \rangle \in f_0$  and  $\langle \langle n, m \rangle, y_0 \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$ . Since it follows that  $a_0 = n = a$  and  $s(a_0) = s(a)$ , theorem 1.4 allows us to assert  $f = f_0$ . Putting it all together, we have  $+(\langle n, m \rangle) = y_0 = c_0 = f_0(b_0) = f_0(m) = f(m)$ . We conclude that  $+(\langle n, s(m) \rangle) = y = s(f(m)) = s(+(\langle n, m \rangle))$ .  $\square$

#### 1.1.4 Multiplication on $\mathbb{N}$

**Theorem 1.6.** *There is a unique binary operation  $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  that satisfies the following two properties for all  $n, m \in \mathbb{N}$ :*

- $n \cdot 1 = n$
- $n \cdot s(m) = (n \cdot m) + n$

*Proof.* We start with uniqueness. Let  $\cdot, \odot$  be arbitrary. Suppose  $\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  and  $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\cdot(\langle n, 1 \rangle) = n \wedge \cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n))$ . Similarly suppose  $\odot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  and  $\forall m \in \mathbb{N} \forall n \in \mathbb{N} (\odot(\langle n, 1 \rangle) = n \wedge \odot(\langle n, s(m) \rangle) = +(\odot(\langle n, m \rangle), n))$ . We want to show that  $\cdot = \odot$ . We do this by defining  $G$  such that

$$\forall z (z \in G \leftrightarrow z \in \mathbb{N} \wedge \forall n \in \mathbb{N} (\cdot(\langle n, z \rangle) = \odot(\langle n, z \rangle)))$$

(where existence is assured by the subset axiom) and showing that  $G = \mathbb{N}$ .

We need to show  $G \subseteq \mathbb{N}$ ,  $1 \in G$ , and  $\forall g \in G (s(g) \in G)$ . The first requirement is straightforward. For the second, we already know  $1 \in \mathbb{N}$ , and see that for arbitrary  $n \in \mathbb{N}$  we have  $\cdot(\langle n, 1 \rangle) = n = \odot(\langle n, 1 \rangle)$ .

For the third requirement let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $\forall n \in \mathbb{N} (\cdot(\langle n, g \rangle) = \odot(\langle n, g \rangle))$ . We know that  $s(g) \in \mathbb{N}$  exists. So since  $\langle n, s(g) \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(\cdot)$ , we have, by our assumptions,  $\cdot(\langle n, s(g) \rangle) = +(\cdot(\langle n, g \rangle), n) = +(\odot(\langle n, g \rangle), n) = \odot(\langle n, s(g) \rangle)$ . So  $s(g) \in G$  and  $G = \mathbb{N}$ .

To prove the desired result, let arbitrary  $z \in \cdot \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ . Then there exists  $a_0, a_1, b \in \mathbb{N}$  such that  $\langle \langle a_0, a_1 \rangle, b \rangle = z \in \cdot$ . So  $b = \cdot(\langle a_0, a_1 \rangle)$ . Since  $a_1 \in \mathbb{N} = G$ , we can then conclude that  $b = \cdot(\langle a_0, a_1 \rangle) = \odot(\langle a_0, a_1 \rangle)$ . So  $z = \langle \langle a_0, a_1 \rangle, b \rangle \in \odot$ . Next suppose  $z \in \odot$ . It then follows from a similar argument that  $z \in \cdot$ . We are done.

(next page)

Now for existence. We start by defining a function  $h$  for all  $q$ , where

$$\forall q \exists h \forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +))$$

(which exists by the subset axiom, where  $\mathbb{N} \times \mathbb{N}$  is a bounding set. Uniqueness for each  $q$  follows from extensionality.) We show that for all  $q \in \mathbb{N}$ , this  $h$  is a function from  $\mathbb{N}$  into  $\mathbb{N}$ . That is

$$\forall q \in \mathbb{N} [\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \rightarrow h : \mathbb{N} \rightarrow \mathbb{N}]$$

Let arbitrary  $q \in \mathbb{N}$ . Suppose that the statement before the implication is true. First to prove functionality. Let  $a \in \text{dom}h$ . Then there exists  $b$  such that  $\langle a, b \rangle \in h$ . Let  $b'$  be arbitrary such that  $\langle a, b' \rangle \in h$ . By our assumptions we must have  $a, b, b' \in \mathbb{N}$ ,  $+(\langle a, q \rangle) = b$ , and  $+(\langle a, q \rangle) = b'$ . Since  $+: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  we must have  $b = b'$ . Next for the domain bound suppose arbitrary  $a \in \text{dom}h$ . Then there exists  $b$  such that  $\langle a, b \rangle \in h$ . It immediately follows that  $a \in \mathbb{N}$ . Suppose instead that  $a \in \mathbb{N}$ . Then  $\langle a, q \rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ . So  $+(\langle a, q \rangle) \in \mathbb{N}$  exists and by definition,  $\langle a, +(\langle a, q \rangle) \rangle \in h$ . So  $a \in \text{dom}h$ . Finally for the range bound, let arbitrary  $b \in \text{ran}h$ . Then there exists  $a$  such that  $\langle a, b \rangle \in h$ . It immediately follows by definition that  $b \in \mathbb{N}$ . We have therefore shown

$$\begin{aligned} \forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \\ \wedge h : \mathbb{N} \rightarrow \mathbb{N}) \end{aligned}$$

Theorem 1.4 states

$$\forall H \forall q \forall h ([q \in H \wedge h : H \rightarrow H] \rightarrow \exists! g (g : \mathbb{N} \rightarrow H \wedge g(1) = q \wedge g \circ s = h \circ g))$$

which allows us to assert

$$\forall q \forall h ([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

(next page)

We showed

$$\forall q \in \mathbb{N} \exists! h (\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, q \rangle, x \rangle \in +)) \wedge h : \mathbb{N} \rightarrow \mathbb{N})$$

and

$$\forall q \forall h ([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

Now we construct  $\cdot$  as the set

$$\begin{aligned} \forall z (z \in \cdot \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N} ( \\ \exists h (\forall z_0 (z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +)) \wedge h : \mathbb{N} \rightarrow \mathbb{N} \wedge \exists g (g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = a \wedge g \circ s = h \circ g \\ \wedge \exists c (\langle b, c \rangle \in g \wedge z = \langle \langle a, b \rangle, c \rangle)))))) \end{aligned}$$

(existence follows from the subset axiom, where  $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$  is a bounding set.)  
We want to verify

$$\cdot : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N} (\cdot(\langle n, 1 \rangle) = n \wedge \cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n))$$

We start with functionality. Let arbitrary  $p \in \text{dom}(\cdot)$ . Then by definition there exists  $q$  such that  $\langle p, q \rangle \in \cdot$ . This means that there exists  $a, b \in \mathbb{N}$  and  $h : \mathbb{N} \rightarrow \mathbb{N}$  such that  $\forall z_0 (z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +))$ . There also exists  $g : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g(1) = a$  and  $g \circ s = h \circ g$ , and  $c$  such that  $\langle b, c \rangle \in g$  and  $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$ . Now suppose there exists  $\langle p, q' \rangle \in \cdot$ . Similarly there exists  $a', b' \in \mathbb{N}$ ,  $h' : \mathbb{N} \rightarrow \mathbb{N}$  such that  $\forall z_0 (z_0 \in h' \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m, x \rangle \wedge \langle \langle m, a' \rangle, x \rangle \in +))$ ,  $g' : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g'(1) = a'$  and  $g' \circ s = h' \circ g'$ , and  $c'$  such that  $\langle b', c' \rangle \in g'$  and  $\langle p, q' \rangle = \langle \langle a', b' \rangle, c' \rangle$ . We have  $\langle a', b' \rangle = p = \langle a, b \rangle$ . So  $a' = a$  and  $b' = b$ . Uniqueness, as proven, implies that  $h = h'$  and  $g = g'$ . So  $q' = c' = g'(b') = g(b') = g(b) = c = q$ .

Next we show  $\text{dom}(\cdot) = \mathbb{N} \times \mathbb{N}$ . Let arbitrary  $p \in \text{dom}(\cdot)$ . Then by definition there exists  $q$  such that  $\langle p, q \rangle \in \cdot$ . So there exists  $a, b \in \mathbb{N}$  and  $c$  such that  $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$ . So  $p = \langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$ . Now let  $p \in \mathbb{N} \times \mathbb{N}$ . Then by definition there exists  $a, b \in \mathbb{N}$  such that  $p = \langle a, b \rangle$ . Since  $a \in \mathbb{N}$ , as proven there exists  $h : \mathbb{N} \rightarrow \mathbb{N}$  such that  $\forall z (z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N} (z = \langle m, x \rangle \wedge \langle \langle m, a \rangle, x \rangle \in +))$ , and there exists  $g : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g(1) = a$  and  $g \circ s = h \circ g$ . By definition we then have  $\langle \langle a, b \rangle, g(b) \rangle \in \cdot$  and  $p = \langle a, b \rangle \in \text{dom}(\cdot)$ .

For the range bound, let  $q \in \text{ran}(\cdot)$ . Then there exists  $p$  such that  $\langle p, q \rangle \in \cdot$ . So there exists  $a, b \in \mathbb{N}$ ,  $g : \mathbb{N} \rightarrow \mathbb{N}$ , and  $c$  such that  $\langle b, c \rangle \in g$  and  $\langle p, q \rangle = \langle \langle a, b \rangle, c \rangle$ . So  $q = c = g(b) \in \mathbb{N}$ .

Finally we verify the two properties of multiplication. Let arbitrary  $m, n \in \mathbb{N}$ . We first show  $\cdot(\langle n, 1 \rangle) = n$ . We know that  $\langle \langle n, 1 \rangle, \cdot(\langle n, 1 \rangle) \rangle \in \cdot$ . By definition there then exists  $a, b \in \mathbb{N}$ ,  $g : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g(1) = a$ , and  $c$  such that  $\langle b, c \rangle \in g$  and  $\langle \langle n, 1 \rangle, \cdot(\langle n, 1 \rangle) \rangle = \langle \langle a, b \rangle, c \rangle$ . So  $\cdot(\langle n, 1 \rangle) = c = g(b) = g(1) = a = n$ .  
(next page)

Now we show  $\cdot(\langle n, s(m) \rangle) = +(\cdot(\langle n, m \rangle), n)$ .  $s(m) \in \mathbb{N}$  exists, so we have  $\langle \langle n, s(m) \rangle, \cdot(\langle n, s(m) \rangle) \rangle \in \cdot$ . Then there exists  $a, b \in \mathbb{N}$ ,  $h : \mathbb{N} \rightarrow \mathbb{N}$  such that  $\forall z_0 (z_0 \in h \leftrightarrow \exists m_0 \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m_0, x \rangle \wedge \langle \langle m_0, a \rangle, x \rangle \in +))$ ,  $g : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g(1) = a$  and  $g \circ s = h \circ g$ , and  $c$  such that  $\langle b, c \rangle \in g$  and  $\langle \langle n, s(m) \rangle, \cdot(\langle n, s(m) \rangle) \rangle = \langle \langle a, b \rangle, c \rangle$ . We then have  $\cdot(\langle n, s(m) \rangle) = c = g(b) = g(s(m)) = h(g(m))$  and  $a = n$ . Similarly we know that  $\langle \langle n, m \rangle, \cdot(\langle n, m \rangle) \rangle \in \cdot$ . So there exists  $a_0, b_0 \in \mathbb{N}$ ,  $h_0 : \mathbb{N} \rightarrow \mathbb{N}$  such that  $\forall z_0 (z_0 \in h_0 \leftrightarrow \exists m_0 \in \mathbb{N} \exists x \in \mathbb{N} (z_0 = \langle m_0, x \rangle \wedge \langle \langle m_0, a_0 \rangle, x \rangle \in +))$ ,  $g_0 : \mathbb{N} \rightarrow \mathbb{N}$  such that  $g_0(1) = a_0$  and  $g_0 \circ s = h_0 \circ g_0$ , and  $c_0$  such that  $\langle b_0, c_0 \rangle \in g_0$  and  $\langle \langle n, m \rangle, \cdot(\langle n, m \rangle) \rangle = \langle \langle a_0, b_0 \rangle, c_0 \rangle$ . This means that  $a_0 = n = a$ , and that by our earlier statements we can assert that  $h = h_0$ , and  $g = g_0$ . So  $\cdot(\langle n, m \rangle) = c_0 = g_0(b_0) = g(m)$ . We then have  $\cdot(\langle n, s(m) \rangle) = h(g(m)) = h(\cdot(\langle n, m \rangle))$ . From the definition of  $h$ , there then exists  $m_0, x \in \mathbb{N}$  such that  $\langle \cdot(\langle n, m \rangle), h(\cdot(\langle n, m \rangle)) \rangle = \langle m_0, x \rangle$  and  $\langle \langle m_0, a \rangle, x \rangle \in +$ . This and the fact that  $a = n$  allows us to conclude  $+(\cdot(\langle n, m \rangle), n) = h(\cdot(\langle n, m \rangle)) = \cdot(\langle n, s(m) \rangle)$ .  $\square$

### 1.1.5 Properties of addition and multiplication on the naturals

**Theorem 1.7.** *Let  $a, b, c \in \mathbb{N}$ .*

1. *If  $a + c = b + c$ , then  $a = b$  (Cancellation Law for Addition)*
2.  *$(a + b) + c = a + (b + c)$  (Associative Law for Addition)*
3.  *$1 + a = s(a) = a + 1$*
4.  *$a + b = b + a$  (Commutative Law for Addition)*
5.  *$a + b \neq 1$*
6.  *$a + b \neq a$*
7.  *$a \cdot 1 = a = 1 \cdot a$  (Identity Law for Multiplication)*
8.  *$(a + b)c = ac + bc$  (Distributive Law)*
9.  *$ab = ba$  (Commutative Law for Multiplication)*
10.  *$c(a + b) = ca + cb$  (Distributive Law)*
11.  *$(ab)c = a(bc)$  (Associative Law for Multiplication)*
12. *If  $ac = bc$  then  $a = b$  (Cancellation Law for Multiplication)*
13.  *$ab = 1$  iff  $a = 1 = b$*

(next page)

*Proof.* (1) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} (a + c = b + c \rightarrow a = b)$$

To do this we define the set  $G$  as the set where

$$\forall c (c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + c = b + c \rightarrow a = b))$$

(where existence follows from the subset axiom using  $\mathbb{N}$  as the bounding set. Uniqueness follows from extensionality.) Then we show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward; we need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $a, b \in \mathbb{N}$ . Suppose that  $a + 1 = b + 1$ . By definition we then have  $s(a) = s(b)$ , so by injectivity  $a = b$  and  $1 \in G$ .

Now the latter. Let arbitrary  $g \in G$ . As defined we then have  $g \in \mathbb{N}$  and  $\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + g = b + g \rightarrow a = b)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $a, b \in \mathbb{N}$  and suppose that  $a + s(g) = b + s(g)$ . Then by the definition of addition we have  $s(a + g) = s(b + g)$  and, by injectivity,  $a + g = b + g$ . Since  $g \in G$  we then have  $a = b$ . So  $G = \mathbb{N}$ . To prove the initial statement, let arbitrary  $c \in \mathbb{N}$ . Then from  $\mathbb{N} = G$  the desired conclusion immediately follows.  $\square$

*Proof.* (2) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} ((a + b) + c = a + (b + c))$$

To do this we define  $G$  as the set where

$$\forall c (c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b) + c = a + (b + c)))$$

(where existence and uniqueness can be easily shown) and show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ .

For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $a, b \in \mathbb{N}$ . By the definition of addition we have

$$(a + b) + 1 = s(a + b) = a + s(b) = a + (b + 1)$$

Now the latter. Let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $\forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b) + g = a + (b + g))$ . We know  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $a, b \in \mathbb{N}$ . By definition we then have

$$(a + b) + s(g) = s((a + b) + g) = s(a + (b + g)) = a + s(b + g) = a + (b + s(g))$$

So  $\mathbb{N} = G$ . To prove the initial statement, let arbitrary  $c \in \mathbb{N} = G$ . The desired conclusion immediately follows.  $\square$

(next page)

*Proof.* (3) We want to show

$$\forall a \in \mathbb{N}(1 + a = s(a) = a + 1)$$

To do this we define  $G$  as the set where

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge 1 + a = s(a) = a + 1)$$

(Showing existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show that  $1 \in G$  and  $\forall g \in G(s(g) \in G)$ . For the former we already know that  $1 \in \mathbb{N}$ , and by definition we have  $1 + 1 = s(1)$ .

Now the latter. Let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $1 + g = s(g) = g + 1$ . We know  $s(g) \in \mathbb{N}$  exists. We then have

$$1 + s(g) = s(1 + g) = s(s(g)) = s(g + 1) = (g + 1) + 1 = s(g) + 1$$

So  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

*Proof.* (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b = b + a)$$

To do this we define  $G$  as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a + b = b + a))$$

(Showing existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show that  $1 \in G$  and  $\forall g \in G(s(g) \in G)$ . For the former we already know  $1 \in \mathbb{N}$ . Letting arbitrary  $b \in \mathbb{N}$ , by (3) we have  $1 + b = b + 1$ .

Now the latter. Let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $\forall b \in \mathbb{N}(g + b = b + g)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $b \in \mathbb{N}$ . Then we have

$$s(g) + b = (g + 1) + b \stackrel{(2)}{=} g + (1 + b) \stackrel{(3)}{=} g + s(b) = s(g + b) = s(b + g) = b + s(g)$$

So  $G = \mathbb{N}$ . Showing the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

(next page)

*Proof.* (5) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + b \neq 1)$$

Let arbitrary  $a, b \in \mathbb{N}$ . Suppose for contradiction that  $a + b = 1$ . We have 2 cases. First suppose  $b = 1$ . Then  $a + b = a + 1 = s(a) = 1$ , and the last equality is a contradiction of the peano postulates. Now suppose  $b \neq 1$ . Then by lemma 1.3 there exists  $x \in \mathbb{N}$  such that  $b = s(x)$ . We then have  $a + b = a + s(x) = s(a + x) = 1$ , which is again a contradiction.  $\square$

*Proof.* (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a + b \neq a)$$

We do this by defining  $G$  as the set such that

$$\forall a (a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N} (a + b \neq a))$$

(Showing existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $b \in \mathbb{N}$ . By (5)  $1 + b \neq 1$ .

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $\forall b \in \mathbb{N} (g + b \neq g)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $b \in \mathbb{N}$ . For contradiction, suppose  $s(g) + b = s(g)$ . We then have

$$s(g) = s(g) + b \stackrel{(4)}{=} b + s(g) = s(b + g)$$

By injectivity of  $s$  and (4) we then have  $g = b + g = g + b$ . This contradicts  $\forall b \in \mathbb{N} (g + b \neq g)$ . So  $G = \mathbb{N}$ . Showing the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

*Proof.* (7) We want to show

$$\forall a \in \mathbb{N} (a \cdot 1 = a = 1 \cdot a)$$

to do this we define  $G$  as the set such that

$$\forall a (a \in G \leftrightarrow a \in \mathbb{N} \wedge a \cdot 1 = a = 1 \cdot a)$$

(where existence and uniqueness can be easily shown.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former we already know  $1 \in \mathbb{N}$ , and  $1 \cdot 1 = 1$ .

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $g \cdot 1 = g = 1 \cdot g$ .  $s(g) \in \mathbb{N}$  exists. We then have

$$s(g) \cdot 1 = s(g) = s(g \cdot 1) = s(1 \cdot g) = (1 \cdot g) + 1 = 1 \cdot s(g)$$

so  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

(next page)

*Proof.* (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} ((a + b)c = (ac) + (bc))$$

To do this we define  $G$  as the set such that

$$\forall c (c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b)c = (ac) + (bc)))$$

(where proving existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $a, b \in \mathbb{N}$ . We have

$$(a + b) \cdot 1 = a + b = (a \cdot 1) + (b \cdot 1)$$

Now the latter. Let arbitrary  $g \in G$ . Then by definition  $g \in \mathbb{N}$  and  $\forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a + b)g = (ag) + (bg))$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $a, b \in \mathbb{N}$ . We have

$$\begin{aligned} (a + b)s(g) &= ((a + b)g) + (a + b) = ((ag) + (bg)) + (a + b) \\ &\stackrel{(2)}{=} (ag) + ((bg) + (a + b)) \stackrel{(4,2)}{=} (ag) + ((bg + b) + a) \\ &= (ag) + ((bs(g)) + a) \stackrel{(4,2)}{=} ((ag) + a) + (bs(g)) \\ &= (as(g)) + (bs(g)) \end{aligned}$$

So  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $c \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

*Proof.* (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (ab = ba)$$

To do this we define  $G$  as the set such that

$$\forall a (a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N} (ab = ba))$$

(where proving existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former, we know  $1 \in \mathbb{N}$ . Let arbitrary  $b \in \mathbb{N}$ . Then by (7) we have  $1 \cdot b = b \cdot 1$ .

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $\forall b \in \mathbb{N} (gb = bg)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $b \in \mathbb{N}$ . We have

$$s(g)b = (g + 1)b \stackrel{(8)}{=} (gb) + (1 \cdot b) \stackrel{(7)}{=} (bg) + b = bs(g)$$

So  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

(next page)



*Proof.* (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} (c(a + b) = (ca) + (cb))$$

Let arbitrary  $a, b, c \in \mathbb{N}$ . We have

$$c(a + b) \stackrel{(9)}{=} (a + b)c \stackrel{(8)}{=} (ac) + (bc) \stackrel{(9)}{=} (ca) + (cb) \quad \square$$

*Proof.* (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} ((ab)c = a(bc))$$

To do this we define  $G$  as the set such that

$$\forall b (b \in G \leftrightarrow b \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall c \in \mathbb{N} ((ab)c = a(bc)))$$

(where proving existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ . For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $a, c \in \mathbb{N}$ . We have

$$(a \cdot 1)c = ac \stackrel{(7)}{=} a(1 \cdot c)$$

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $\forall a \in \mathbb{N} \forall c \in \mathbb{N} ((ag)c = a(gc))$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $a, c \in \mathbb{N}$ . We have

$$\begin{aligned} (as(g))c &= ((ag) + a)c \stackrel{(8)}{=} ((ag)c) + (ac) = (a(gc)) + (ac) \\ &\stackrel{(10)}{=} a((gc) + c) \stackrel{(9)}{=} a((cg) + c) = a(cs(g)) \\ &\stackrel{(9)}{=} a(s(g)c) \end{aligned}$$

So  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $b \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

(next page)

*Proof.* (12) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} (ac = bc \rightarrow a = b)$$

To do this we define  $G$  as the set such that

$$\forall a (a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N} \forall c \in \mathbb{N} (ac = bc \rightarrow a = b))$$

(where proving existence and uniqueness is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in G$  and  $\forall g \in G (s(g) \in G)$ .

For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $b, c \in \mathbb{N}$  and suppose that  $1 \cdot c = bc$ . Suppose for contradiction that  $b \neq 1$ . Then by lemma 1.3 there exists  $x \in \mathbb{N}$  such that  $b = s(x)$ . We then have

$$c \stackrel{(7)}{=} 1 \cdot c = bc = s(x)c = (x+1)c \stackrel{(8,7)}{=} (xc) + c \stackrel{(4,9)}{=} c + (cx)$$

which contradicts (6).

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $\forall b \in \mathbb{N} \forall c \in \mathbb{N} (gc = bc \rightarrow g = b)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $b, c \in \mathbb{N}$ . Suppose  $s(g)c = bc$ . For contradiction, suppose  $b = 1$ . Then

$$c \stackrel{(7)}{=} 1 \cdot c = bc = s(g)c \stackrel{(9)}{=} cs(g) = (cg) + c \stackrel{(4)}{=} c + (cg)$$

contradicting (6). So we must have  $b \neq 1$  and by lemma 1.3 there exists  $x \in \mathbb{N}$  such that  $b = s(x)$ . Then

$$(cg) + c = cs(g) \stackrel{(9)}{=} s(g)c = bc = s(x)c = (x+1)c \stackrel{(8,7)}{=} (xc) + c$$

and by (1),  $gc = cg = xc$ . Since  $g \in G$  we then have  $g = x$  and  $s(g) = s(x) = b$ . So  $G = \mathbb{N}$ . Proving the initial statement is straightforward. Let arbitrary  $a \in \mathbb{N} = G$  and the desired conclusion immediately follows.  $\square$

*Proof.* (13) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (ab = 1 \leftrightarrow a = 1 = b)$$

Let arbitrary  $a, b \in \mathbb{N}$ . First suppose  $ab = 1$ . For contradiction suppose  $b \neq 1$ . Then by lemma 1.3 there exists  $x \in \mathbb{N}$  such that  $b = s(x)$ . So

$$1 = ab = as(x) = a(x+1) \stackrel{(10)}{=} (ax) + (a \cdot 1) = (ax) + a$$

which contradicts (5). So  $b = 1$ . We immediately have  $1 = ab = a \cdot 1 = a$  and  $a = 1 = b$ . Now suppose  $a = 1 = b$ . We immediately have  $ab = 1 \cdot 1 = 1$ .  $\square$

### 1.1.6 Ordering on $\mathbb{N}$

**Definition 1.8.** The relation  $<$  on  $\mathbb{N}$  is defined by  $a < b$  if and only if there is some  $p \in \mathbb{N}$  such that  $a + p = b$  for all  $a, b \in \mathbb{N}$ .

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a < b \leftrightarrow \exists p \in \mathbb{N} (a + p = b))$$

The relation  $\leq$  on  $\mathbb{N}$  is defined by  $a \leq b$  if and only if  $a < b$  or  $a = b$  for all  $a, b \in \mathbb{N}$ .

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a \leq b \leftrightarrow (a < b \vee a = b))$$

We will write  $a > b$  to mean the same thing as  $b < a$ . The same applies to  $a \geq b$ .

**Theorem 1.9.** Let  $a, b \in \mathbb{N}$ . Suppose that  $a < b$ . Then there is a unique  $p \in \mathbb{N}$  such that  $a + p = b$ .

*Proof.* By definition there exists  $p \in \mathbb{N}$  such that  $a + p = b$ . Let  $p' \in \mathbb{N}$  be arbitrary such that  $a + p' = b$ . Then by part 4 of theorem 1.7 we have  $p + a = b$  and  $p' + a = b$ , so  $p + a = p' + a$ . By part 1 of theorem 1.7 we have  $p = p'$ .  $\square$

**Theorem 1.10.** Let  $a, b, c \in \mathbb{N}$ .

1.  $a \leq a$ , and  $a \not< a$ , and  $a < a + 1$
2.  $1 \leq a$
3. If  $a < b$  and  $b < c$ , then  $a < c$ ; if  $a \leq b$  and  $b < c$ , then  $a < c$ ; if  $a < b$  and  $b \leq c$ , then  $a < c$ ; if  $a \leq b$  and  $b \leq c$ , then  $a \leq c$
4.  $a < b$  if and only if  $a + c < b + c$
5. Precisely one of  $a < b$  or  $a = b$  or  $a > b$  holds (Trichotomy Law)
6.  $a < b$  if and only if  $ac < bc$
7.  $a \leq b$  or  $b \leq a$
8. If  $a \leq b$  and  $b \leq a$ , then  $a = b$
9. It cannot be that  $b < a < b + 1$
10.  $a \leq b$  if and only if  $a < b + 1$
11.  $a < b$  if and only if  $a + 1 \leq b$

(next page)

*Proof.* (1) We want to show

$$\forall a \in \mathbb{N}(a \leq a \wedge a \not\leq a \wedge a < a + 1)$$

Let arbitrary  $a \in \mathbb{N}$ . Since  $a = a$ , by definition  $a \leq a$ . Next suppose for contradiction that  $a < a$ . Then there would exist  $p \in \mathbb{N}$  such that  $a + p = a$ , contradicting part 6 of theorem 1.7. Finally since  $1 \in \mathbb{N}$  and  $a + 1 = a + 1$ , by definition  $a < a + 1$ .  $\square$

*Proof.* (2) We want to show

$$\forall a \in \mathbb{N}(1 \leq a)$$

Let arbitrary  $a \in \mathbb{N}$ . We have 2 cases. First suppose  $a = 1$ ; then by definition  $1 \leq a$ . Next suppose  $a \neq 1$ . Then by lemma 1.3 there exists  $x \in \mathbb{N}$  such that  $a = s(x) = x + 1$ . By theorem 1.7 part 4 we have  $a = 1 + x$ . So by definition  $1 < a$ .  $\square$

*Proof.* (3) We want to show

$$\begin{aligned} \forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N} ([ (a < b \wedge b < c) \rightarrow a < c ] \wedge [ (a \leq b \wedge b < c) \rightarrow a < c ] \\ \wedge [ (a < b \wedge b \leq c) \rightarrow a < c ] \wedge [ (a \leq b \wedge b \leq c) \rightarrow a \leq c ]) \end{aligned}$$

Let arbitrary  $a, b, c \in \mathbb{N}$ . First suppose  $a < b \wedge b < c$ . Then there exists  $x, y \in \mathbb{N}$  such that  $a + x = b$  and  $b + y = c$ . We then have  $(a + x) + y = b + y = c$ . By theorem 1.7 part 2 we have  $a + (x + y) = c$  and therefore  $a < c$ .

Next suppose  $a \leq b \wedge b < c$ . There exists  $y \in \mathbb{N}$  such that  $b + y = c$ . We have 2 cases. First suppose  $a < b$ . Then as proven earlier we have  $a < c$ . Next suppose  $a = b$ . Then we immediately have  $a + y = c$  and  $a < c$ .

Next suppose  $a < b \wedge b \leq c$ . There exists  $x \in \mathbb{N}$  such that  $a + x = b$ . We have 2 cases. First suppose  $b < c$ . Then again we immediately have  $a < c$ . Next suppose  $b = c$ . Then  $a + x = c$  and  $a < c$ .

Finally suppose  $a \leq b \wedge b \leq c$ . We have 4 cases. First suppose  $a < b \wedge b < c$ . As proven  $a < c$ . Next suppose  $a = b \wedge b < c$ . As proven  $a < c$ . Next suppose  $a < b \wedge b = c$ . As proven  $a < c$ . Finally suppose  $a = b$  and  $b = c$ . It immediately follows that  $a = c$  and  $a \leq c$ .  $\square$

*Proof.* (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(a < b \leftrightarrow a + c < b + c)$$

Let arbitrary  $a, b, c \in \mathbb{N}$ . First suppose  $a < b$ . Then there exists  $x \in \mathbb{N}$  such that  $a + x = b$ . Then  $(a + x) + c = b + c$  and by theorem 1.7 parts 2 and 4 we have  $(a + c) + x = b + c$ . So  $a + c < b + c$ .

Next suppose  $a + c < b + c$ . Then there exists  $x \in \mathbb{N}$  such that  $(a + c) + x = b + c$ . By theorem 1.7 parts 2 and 4 we have  $(a + x) + c = b + c$ , and by part 1 of that same theorem we have  $a + x = b$ . So  $a < b$ .  $\square$

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*Proof.* (5) We want to prove trichotomy

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a < b \vee a = b \vee b < a) \\ \wedge \neg((a < b) \wedge (a = b)) \wedge \neg((a < b) \wedge (b < a)) \wedge \neg((a = b) \wedge (b < a)))$$

Let arbitrary  $a, b \in \mathbb{N}$ . We start by proving the last three statements. First suppose for contradiction that  $a < b$  and  $a = b$ . Then  $a < a$ , contradicting (1). Next suppose for contradiction that  $a = b$  and  $b < a$ . A similar argument shows that this contradicts (1). Finally suppose for contradiction that  $a < b \wedge b < a$ . By (3) we have  $a < a$ , a contradiction.

Now for the first statement. We define  $G$  as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a < b \vee a = b \vee b < a))$$

(proving existence and uniqueness of  $G$  is straightforward.) We show that  $G = \mathbb{N}$ . Showing  $G \subseteq \mathbb{N}$  is straightforward. We need to show that  $1 \in G$  and  $\forall g \in G(s(g) \in G)$ . For the former, we already know  $1 \in \mathbb{N}$ . Let arbitrary  $b \in \mathbb{N}$ . By (2) we have  $1 \leq b$ , meaning  $1 < b$  or  $1 = b$ . Either case allows us to conclude that  $1 \in G$ .

Now the latter. Let arbitrary  $g \in G$ . Then  $g \in \mathbb{N}$  and  $\forall b \in \mathbb{N}(g < b \vee g = b \vee b < g)$ .  $s(g) \in \mathbb{N}$  exists. Let arbitrary  $b \in \mathbb{N}$ . We have 3 cases. First suppose  $g < b$ . Then there exists  $x \in \mathbb{N}$  such that  $g + x = b$ . Suppose  $x = 1$ ; then  $s(g) = g + 1 = b$ . Next suppose  $x \neq 1$ ; then by lemma 1.3 there exists  $y \in \mathbb{N}$  such that  $x = s(y)$ . So  $b = g + x = g + s(y) = g + (y + 1)$ . By theorem 1.7 parts 2 and 4 we have  $(g + 1) + y = b$  and  $s(g) < b$ . For the second case suppose  $g = b$ . Then by (1) we have  $b = g < g + 1 = s(g)$ . Finally suppose  $b < g$ . Then since  $g < g + 1$  by (3) we have  $b < g + 1 = s(g)$ . We then have  $G = \mathbb{N}$ .

Showing the desired result is straightforward. For all arbitrary  $a \in \mathbb{N}$ ,  $a \in G$ . The result immediately follows.  $\square$

*Proof.* (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac < bc \leftrightarrow a < b)$$

Let arbitrary  $a, b, c \in \mathbb{N}$ . First suppose  $a < b$ . Then there exists  $x \in \mathbb{N}$  such that  $a + x = b$ . Then by theorem 1.7 part 8 we have  $bc = (a + x)c = (ac) + (xc)$ ; so  $ac < bc$ .

Next suppose  $ac < bc$ . Suppose for contradiction that  $a \not< b$ . Then by (5) we have  $a = b \vee b < a$ . First suppose  $a = b$ . Then we immediately have  $ac < ac$ , which contradicts (1). Next suppose  $b < a$ . There exists  $x, y \in \mathbb{N}$  such that  $(ac) + x = bc$  and  $b + y = a$ . By theorem 1.7 parts 8 and 2 we have  $bc = ((b + y)c) + x = ((bc) + (yc)) + x = (bc) + ((yc) + x)$ , contradicting part 6 of that same theorem.  $\square$

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*Proof.* (7) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a \leq b \vee b \leq a)$$

Let arbitrary  $a, b \in \mathbb{N}$ . First suppose  $a \not\leq b$ . Then by (5)  $a = b \vee b < a$ , meaning  $b \leq a$ . Next suppose  $a < b$ . Then immediately it follows that  $a \leq b$ .  $\square$

*Proof.* (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} ((a \leq b \wedge b \leq a) \rightarrow a = b)$$

Let arbitrary  $a, b \in \mathbb{N}$ . Suppose that  $a \leq b \wedge b \leq a$ . We have four cases. Three of the cases:  $a < b \wedge b < a$ ,  $a = b \wedge b < a$ , and  $a < b \wedge b = a$  all contradict (5). The only possibility is the last case  $a = b \wedge b = a$ .  $\square$

*Proof.* (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (\neg(b < a \wedge a < b + 1))$$

Let arbitrary  $a, b \in \mathbb{N}$ . Suppose for contradiction that  $b < a \wedge a < b + 1$ . Then there exists  $x, y \in \mathbb{N}$  such that  $b + x = a$  and  $a + y = b + 1$ . Then  $(b + x) + y = b + 1$ . By theorem 1.7 parts 2 and 4 we have  $(x + y) + b = 1 + b$ . By part 1 of that theorem we then have  $x + y = 1$ , contradicting part 5 of that theorem.  $\square$

*Proof.* (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a \leq b \leftrightarrow a < b + 1)$$

Let arbitrary  $a, b \in \mathbb{N}$ . First suppose  $a \leq b$ . Suppose for contradiction that  $b + 1 \leq a$ . Then by (3) we would have  $b + 1 \leq b$ . The case where  $b + 1 = b$  is a contradiction to theorem 1.7 part 6. The case where  $b + 1 < b$  implies the existence of  $x \in \mathbb{N}$  such that  $(b + 1) + x = b$ ; by theorem 1.7 part 2 we have  $b + (1 + x) = b$ , again contradicting part 6 of that theorem. We can then conclude that  $\neg(b + 1 \leq a)$ , meaning  $\neg(b + 1 = a \vee b + 1 < a)$ . By (5) we must then have  $a < b + 1$ .

Now for the other direction. Suppose  $a < b + 1$ . Supposing that  $b < a$  immediately contradicts (9). Therefore by (5) we must have  $a \leq b$ .  $\square$

*Proof.* (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} (a < b \leftrightarrow a + 1 \leq b)$$

Let arbitrary  $a, b \in \mathbb{N}$ . Suppose  $a < b$ . Supposing  $b < a + 1$  immediately contradicts (9), so we must have, by (5),  $a + 1 \leq b$ .

Next suppose  $a + 1 \leq b$ . Suppose for contradiction that  $b \leq a$ , which would imply by (3) that  $a + 1 \leq a$ . The case where  $a + 1 = a$  immediately contradicts theorem 1.7 part 6. The case where  $a + 1 < a$  implies the existence of  $x \in \mathbb{N}$  such that  $(a + 1) + x = a$ ; theorem 1.7 part 2 then implies  $a + (1 + x) = a$ , again contradicting part 6 of that same theorem. We conclude that  $\neg(b = a \vee b < a)$ , so by (5) we have  $a < b$ .  $\square$

### 1.1.7 Well-ordering

**Theorem 1.11.** (*Well-Ordering Principle*) Let  $G \subseteq \mathbb{N}$  be a non-empty set. Then there is some  $m \in G$  such that  $m \leq g$  for all  $g \in G$ .

*Proof.* We want to show

$$\forall G(G \subseteq \mathbb{N} \wedge G \neq \emptyset \rightarrow \exists m \in G(\forall g \in G(m \leq g)))$$

Let  $G$  be arbitrary; assume that  $G \subseteq \mathbb{N}$  and  $G \neq \emptyset$ . Assume for contradiction that the implication is false. That is

$$\neg \exists m \in G(\forall g \in G(m \leq g))$$

We define  $H$  as the set such that

$$\forall a(a \in H \leftrightarrow a \in \mathbb{N} \wedge \forall n \in \mathbb{N}(n \leq a \rightarrow n \notin G))$$

(where existence and uniqueness can be easily shown). We can show that  $H \cap G = \emptyset$ . For contradiction suppose there exists  $x$  such that  $x \in H$  and  $x \in G$ . By definition then  $x \in \mathbb{N}$  and  $\forall n \in \mathbb{N}(n \leq x \rightarrow n \notin G)$ . Then since  $x \leq x$  we must have  $x \notin G$ , a contradiction. We will show that  $H = \mathbb{N}$ . Showing  $H \subseteq \mathbb{N}$  is straightforward. We need to show  $1 \in H$  and  $\forall h \in H(s(h) \in H)$ .

We start with the former. Suppose for contradiction that  $1 \notin H$ . We know  $1 \in \mathbb{N}$ , so we must have  $\neg \forall n \in \mathbb{N}(n \leq 1 \rightarrow n \notin G)$ . This is logically equivalent to

$$\exists n \in \mathbb{N}(n \leq 1 \wedge n \in G)$$

That is, there exists  $n \in \mathbb{N}$  such that  $n \leq 1$  and  $n \in G$ .  $n < 1$  is a contradiction as it would imply the existence of  $x \in \mathbb{N}$  such that  $n + x = 1$ , contradicting theorem 1.7 part 5. We therefore must have  $1 = n \in G$ . Let arbitrary  $g \in G \subseteq \mathbb{N}$ . By theorem 1.10 part 2 we must have  $n = 1 \leq g$ . This contradicts our assumption that such an element does not exist, so  $1 \in H$ .

Now the latter. Let arbitrary  $h \in H$ . Then  $h \in \mathbb{N}$  and  $\forall n \in \mathbb{N}(n \leq h \rightarrow n \notin G)$ . We know  $s(h) \in \mathbb{N}$  exists. Suppose for contradiction that  $s(h) \notin H$ . Then as before there then exists  $n \in \mathbb{N}$  such that  $n \leq s(h)$  and  $n \in G$ . By the contrapositive of one of the consequences of  $h \in H$ , since  $n \in G$  we then have  $\neg(n \leq h)$ , which by theorem 1.10 part 5 implies  $h < n$ . Since  $n < s(h) = h + 1$  would then lead to a contradiction of theorem 1.10 part 9, we must have  $n = s(h)$ . Let arbitrary  $g \in G$ . Suppose for contradiction that  $g < n = s(h) = h + 1$ . Then by theorem 1.10 part 10 we have  $g \leq h$ . Since  $h \in H$  we then have  $g \notin G$ , a contradiction. So we must have  $\neg(g < n)$ , which by theorem 1.10 part 5 implies  $n \leq g$ . We know that  $n \in G$ , so this contradicts our earlier assumption that such an element must not exist. So  $s(h) \in H$  and  $H = \mathbb{N}$ .

We know  $H \cap G = \emptyset$ . So  $\mathbb{N} \cap G = \emptyset$ . Suppose for contradiction that  $G$  is nonempty, meaning that there exists some  $x \in G$ . Then since  $G \subseteq \mathbb{N}$  we would have  $x \in \mathbb{N}$  and therefore  $x \in \mathbb{N} \cap G = \emptyset$ , a contradiction. So we must have  $G = \emptyset$ , contradicting a key assumption.  $\square$

## 1.2 Construction of the integers

### 1.2.1 Construction

**Definition 1.12.** The relation  $\sim$  on  $\mathbb{N} \times \mathbb{N}$  is defined by  $\langle a, b \rangle \sim \langle c, d \rangle$  if and only if  $a + d = b + c$  for all  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$ . That is, the set such that

$$\forall x (x \in \sim \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} (x = \langle \langle a, b \rangle, \langle c, d \rangle \rangle \wedge a + d = b + c))$$

(Existence follows from using  $(\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$  as a bounding set. Uniqueness from extensionality)

**Lemma 1.13.** The relation  $\sim$  is an equivalence relation on  $\mathbb{N} \times \mathbb{N}$ .

*Proof.* Showing that  $\sim \subseteq (\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$  is straightforward. We need to show reflexivity on  $\mathbb{N} \times \mathbb{N}$ , symmetry, and transitivity. We start with reflexivity; let arbitrary  $\langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$ . We immediately have  $a + b = b + a$  and  $\langle a, b \rangle \sim \langle a, b \rangle$ . For symmetry, let  $\langle a, b \rangle, \langle c, d \rangle$  be arbitrary and suppose  $\langle a, b \rangle \sim \langle c, d \rangle$ . Then by definition both pairs are members of  $\mathbb{N} \times \mathbb{N}$  and  $a + d = b + c$ . We then have  $c + b = b + c = a + d = d + a$ , which then means  $\langle c, d \rangle \sim \langle a, b \rangle$ .

Finally transitivity. Let  $\langle a, b \rangle, \langle c, d \rangle, \langle e, f \rangle$  be arbitrary, suppose  $\langle a, b \rangle \sim \langle c, d \rangle$  and  $\langle c, d \rangle \sim \langle e, f \rangle$ . This means that all three pairs are members of  $\mathbb{N} \times \mathbb{N}$ ,  $a + d = c + b$ , and  $c + f = e + d$ . We then have

$$\begin{aligned} (a + f) + c &= a + (c + f) = a + (e + d) = (a + d) + e \\ &= (c + b) + e = (b + e) + c \end{aligned}$$

By cancellation we then have  $a + f = b + e$ . So  $\langle a, b \rangle \sim \langle e, f \rangle$ .  $\square$

**Definition 1.14.** The set of *integers*, denoted  $\mathbb{Z}$ , is the quotient set  $(\mathbb{N} \times \mathbb{N}) / \sim$ . That is,

$$\forall x (x \in \mathbb{Z} \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} (x = [\langle a, b \rangle]))$$

where

$$\forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall x (x \in [\langle a, b \rangle] \leftrightarrow \langle a, b \rangle \sim x)$$

Each integer is an equivalence class. Existence of each equivalence class can be shown using  $\text{ran}(\sim)$  as a bounding set. The same can be done for  $\mathbb{Z}$  using  $\mathcal{P}(\text{ran}(\sim))$ . Uniqueness for both follows from extensionality.

**Definition 1.15.** We define  $\hat{0}, \hat{1}$  as  $\hat{0} = [1, 1]$  and  $\hat{1} = [1 + 1, 1]$ .



## 1.2.2 Operations on the integers

**Definition 1.16.** The binary operation  $+$  on  $\mathbb{Z}$  is defined as  $+: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$  where

$$\forall x(x \in + \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} (x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle, [\langle a+c, b+d \rangle]))$$

put simply, for all  $[\langle a, b \rangle], [\langle c, d \rangle] \in \mathbb{Z}$ ,

$$[\langle a, b \rangle] + [\langle c, d \rangle] = [\langle a+c, b+d \rangle]$$

(A bounding set is  $(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{Z}$ . Uniqueness follows from extensionality.)

*Proof.* We start by showing the domain bound  $\text{dom}(+) = \mathbb{Z} \times \mathbb{Z}$ . First suppose  $x \in \text{dom}(+)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in +$  and  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \in \mathbb{Z} \times \mathbb{Z}$ . Next suppose  $x \in \mathbb{Z} \times \mathbb{Z}$ . Then there exists  $x_0, x_1 \in \mathbb{Z}$  such that  $x = \langle x_0, x_1 \rangle$ . There exists  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x_0 = [\langle a, b \rangle]$  and  $x_1 = [\langle c, d \rangle]$ . We know  $\langle a+c, b+d \rangle \in \mathbb{N} \times \mathbb{N}$  and  $\langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle, [\langle a+c, b+d \rangle] \rangle \in +$ , so  $x = \langle x_0, x_1 \rangle = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \in \text{dom}(+)$ .

Now the range bound  $\text{ran}(+) \subseteq \mathbb{Z}$ . Let arbitrary  $y \in \text{ran}(+)$ . Then there exists  $x$  such that  $\langle x, y \rangle \in +$  and  $a, b, c, d \in \mathbb{N}$  such that  $y = [\langle a+c, b+d \rangle] \in \mathbb{Z}$ .

Now we prove well definedness. That is, that addition between equivalence classes is independent of the particular elements used to represent each equivalence class:

$$\begin{aligned} \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N} \\ [\langle a, b \rangle] = [\langle a', b' \rangle] \wedge [\langle c, d \rangle] = [\langle c', d' \rangle] \\ \rightarrow [\langle a+c, b+d \rangle] = [\langle a'+c', b'+d' \rangle] \end{aligned}$$

Let arbitrary  $\langle a, b \rangle, \langle c, d \rangle, \langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$  and suppose that  $[\langle a, b \rangle] = [\langle a', b' \rangle]$  and  $[\langle c, d \rangle] = [\langle c', d' \rangle]$ . Then  $\langle a, b \rangle \sim \langle a', b' \rangle$  and  $a+b' = b+a'$ . Similarly  $c+d' = d+c'$ . We then have, by associativity and commutativity of natural arithmetic:

$$\begin{aligned} (a+c) + (b'+d') &= (a+b') + (c+d') \\ &= (b+a') + (d+c') = (b+d) + (a'+c') \end{aligned}$$

and we can conclude that  $\langle a+c, b+d \rangle \sim \langle a'+c', b'+d' \rangle$  and  $[\langle a+c, b+d \rangle] = [\langle a'+c', b'+d' \rangle]$ .

Finally we show functionality. Let arbitrary  $x \in \text{dom}(+)$ . Then by definition there exists  $y$  such that  $\langle x, y \rangle \in +$ . There then exists  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle$  and  $y = [\langle a+c, b+d \rangle]$ . Let  $y'$  be arbitrary such that  $\langle x, y' \rangle \in +$ . Then similarly there exists  $\langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a', b' \rangle], [\langle c', d' \rangle] \rangle$  and  $y' = [\langle a'+c', b'+d' \rangle]$ . We know  $\langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle = \langle [\langle a', b' \rangle], [\langle c', d' \rangle] \rangle$ . So since addition is well defined we have  $y = [\langle a+c, b+d \rangle] = [\langle a'+c', b'+d' \rangle] = y'$ .  $\square$

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**Definition 1.17.** The binary operation  $\cdot$  on  $\mathbb{Z}$  is defined as  $\cdot : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$  where

$$\forall x(x \in \cdot \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} ( \\ x = \langle \langle [a, b], [c, d] \rangle, [ac + bd, ad + bc] \rangle))$$

(A bounding set is  $(\mathbb{Z} \times \mathbb{Z}) \times \mathbb{Z}$ . Uniqueness follows from extensionality.)

*Proof.* We start with the domain bound  $\text{dom}(\cdot) = \mathbb{Z} \times \mathbb{Z}$ . First suppose  $x \in \text{dom}(\cdot)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in \cdot$ , and  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle \langle [a, b], [c, d] \rangle, [ac + bd, ad + bc] \rangle \in \mathbb{Z} \times \mathbb{Z}$ . Next suppose  $x \in \mathbb{Z} \times \mathbb{Z}$ . Then there exists  $x_0, x_1 \in \mathbb{Z}$  such that  $x = \langle x_0, x_1 \rangle$ . There exists  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x_0 = [a, b]$  and  $x_1 = [c, d]$ . We have  $\langle \langle [a, b], [c, d] \rangle, [ac + bd, ad + bc] \rangle \in \cdot$ . So  $x = \langle x_0, x_1 \rangle = \langle \langle [a, b], [c, d] \rangle, [ac + bd, ad + bc] \rangle \in \text{dom}(\cdot)$ .

Next the range bound  $\text{ran}(\cdot) \subseteq \mathbb{Z}$ . Let arbitrary  $y \in \text{ran}(\cdot)$ . Then there exists  $x$  such that  $\langle x, y \rangle \in \cdot$  and  $a, b, c, d \in \mathbb{N}$  such that  $y = [ac + bd, ad + bc] \in \mathbb{Z}$ .

Next we show well definedness. That is

$$\begin{aligned} \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N} ( \\ ([a, b] = [a', b'] \wedge [c, d] = [c', d']) \\ \rightarrow [ac + bd, ad + bc] = [a'c' + b'd', a'd' + b'c']) \end{aligned}$$

Let arbitrary  $\langle a, b \rangle, \langle c, d \rangle, \langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$  and suppose  $[a, b] = [a', b']$  and  $[c, d] = [c', d']$ . Then we have  $\langle a, b \rangle \sim \langle a', b' \rangle$  and  $a + b' = b + a'$ . Similarly we have  $c + d' = d + c'$ . We want to show

$$(ac + bd) + (a'd' + b'c') = (ad + bc) + (a'c' + b'd')$$

See that

$$\begin{aligned} & ((a + b')d + (b + a')c) + ((c + d')b' + (d + c')a') \\ &= ((ad + b'd) + (bc + a'c)) + ((cb' + d'b') + (da' + c'a')) \\ &= ((ad + bc) + (b'd + a'c)) + ((a'c' + b'd') + (da' + cb')) \\ &= ((ad + bc) + (a'c' + b'd')) + ((b'd + a'c) + (da' + cb')) \end{aligned}$$

Also see that

$$\begin{aligned} & ((a + b')d + (b + a')c) + ((c + d')b' + (d + c')a') \\ &= ((b + a')d + (a + b')c) + ((d + c')b' + (c + d')a') \\ &= ((bd + a'd) + (ac + b'c)) + ((db' + c'b') + (ca' + d'a')) \\ &= ((ac + bd) + (da' + cb')) + ((a'd' + c'b') + (a'c + b'd)) \\ &= ((ac + bd) + (a'd' + c'b')) + ((b'd + a'c) + (da' + cb')) \end{aligned}$$

By cancellation we then have our desired result, which means

$$\langle ac + bd, ad + bc \rangle \sim \langle a'c' + b'd', a'd' + b'c' \rangle$$

and  $[ac + bd, ad + bc] = [a'c' + b'd', a'd' + b'c']$ .

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Finally functionality. Let arbitrary  $x \in \text{dom}(\cdot)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in \cdot$  and  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle$  and  $y = [\langle ac + bd, ad + bc \rangle]$ . Let  $y'$  be arbitrary such that  $\langle x, y' \rangle \in \cdot$ . Then similarly there exists  $\langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a', b' \rangle], [\langle c', d' \rangle] \rangle$  and  $y' = [\langle a'c' + b'd', a'd' + b'c' \rangle]$ . We have  $\langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle = \langle [\langle a', b' \rangle], [\langle c', d' \rangle] \rangle$ , by well definedness we then have  $y = [\langle ac + bd, ad + bc \rangle] = [\langle a'c' + b'd', a'd' + b'c' \rangle] = y'$ .  $\square$

**Definition 1.18.** The unary operation  $-$  on  $\mathbb{Z}$  is defined as  $- : \mathbb{Z} \rightarrow \mathbb{Z}$  where

$$\forall x(x \in - \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N}(x = \langle [\langle a, b \rangle], [\langle b, a \rangle] \rangle))$$

(A bounding set is  $\mathbb{Z} \times \mathbb{Z}$ . Uniqueness follows from extensionality.)

*Proof.* We start with the domain bound  $\text{dom}(-) = \mathbb{Z}$ . First let arbitrary  $x \in \text{dom}(-)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in -$ , and  $\langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = [\langle a, b \rangle] \in \mathbb{Z}$ . Next suppose  $x \in \mathbb{Z}$ . Then there exists  $\langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ . We have  $\langle [\langle a, b \rangle], [\langle b, a \rangle] \rangle \in -$ . So  $x = [\langle a, b \rangle] \in \text{dom}(-)$ .

Next the range bound  $\text{ran}(-) \subseteq \mathbb{Z}$ . Suppose  $y \in \text{ran}(-)$ . Then there exists  $x$  such that  $\langle x, y \rangle \in -$  and  $a, b \in \mathbb{N}$  such that  $y = [\langle b, a \rangle] \in \mathbb{Z}$ .

Next we show well-definedness. That is

$$\forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N}([\langle a, b \rangle] = [\langle a', b' \rangle] \rightarrow [\langle b, a \rangle] = [\langle b', a' \rangle])$$

Letting arbitrary  $\langle a, b \rangle, \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N}$  and supposing that  $[\langle a, b \rangle] = [\langle a', b' \rangle]$ , we have  $a + b' = b + a'$ . See that this also means  $\langle b, a \rangle \sim \langle b', a' \rangle$  and therefore  $[\langle b, a \rangle] = [\langle b', a' \rangle]$ .

Finally functionality. Let arbitrary  $x \in \text{dom}(-)$ . Then there exists  $y$  such that  $\langle x, y \rangle \in -$ , and  $\langle a, b \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = [\langle a, b \rangle]$  and  $y = [\langle b, a \rangle]$ . Let  $y'$  be arbitrary such that  $\langle x, y' \rangle \in -$ . Then there exists  $\langle a', b' \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = [\langle a', b' \rangle]$  and  $y' = [\langle b', a' \rangle]$ . We have  $[\langle a, b \rangle] = [\langle a', b' \rangle]$ ; by well-definedness we have  $y = [\langle b, a \rangle] = [\langle b', a' \rangle] = y'$ .  $\square$

### 1.2.3 Ordering

**Definition 1.19.** *The binary relation  $<$  on  $\mathbb{Z}$  is defined such that*

$$\forall x(x < \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} (x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \wedge a + d < b + c))$$

( $\mathbb{Z} \times \mathbb{Z}$  is a bounding set. Uniqueness follows from extensionality.)

**Lemma 1.20.**

$$\begin{aligned} \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N} \\ ([\langle a, b \rangle] = [\langle a', b' \rangle] \wedge [\langle c, d \rangle] = [\langle c', d' \rangle]) \\ \rightarrow a + d < b + c \leftrightarrow a' + d' < b' + c' \end{aligned}$$

*Proof.* Let arbitrary  $\langle a, b \rangle, \langle c, d \rangle, \langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$ . Suppose that  $[\langle a, b \rangle] = [\langle a', b' \rangle]$  and  $[\langle c, d \rangle] = [\langle c', d' \rangle]$ , which then means  $a + b' = b + a'$  and  $c + d' = d + c'$ . First suppose  $a + d < b + c$ . Then there exists  $x$  such that  $(a + d) + x = b + c$ , and we have

$$(((a + d) + x) + (b + a')) + (c + d') = ((b + c) + (a + b')) + (d + c')$$

Rearranging the left side:

$$\begin{aligned} & (((a + d) + x) + (b + a')) + (c + d') \\ &= ((a + d) + x) + ((b + c) + (a' + d')) \\ &= (a + d) + (x + ((b + c) + (a' + d'))) \\ &= (a + d) + ((b + c) + (x + (a' + d'))) \\ &= ((a + d) + (b + c)) + ((a' + d') + x) \end{aligned}$$

and the right side:

$$\begin{aligned} & ((b + c) + (a + b')) + (d + c') \\ &= (b + c) + ((a + b') + (d + c')) \\ &= (b + c) + ((a + d) + (b' + c')) \\ &= ((a + d) + (b + c)) + (b' + c') \end{aligned}$$

By cancellation we have  $(a' + d') + x = b' + c'$  and  $a' + d' < b' + c'$ .

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Next suppose  $a' + d' < b' + c'$ . Then there exists  $x$  such that  $a' + d' + x = b' + c'$ . We have

$$(((a' + d') + x) + (a + b')) + (d + c') = ((b' + c') + (b + a')) + (c + d')$$

Rearranging the left side:

$$\begin{aligned} & (((a' + d') + x) + (a + b')) + (d + c') \\ &= ((a' + d') + x) + ((a + b') + (d + c')) \\ &= ((a' + d') + x) + ((b' + c') + (a + d)) \\ &= (a' + d') + (x + ((b' + c') + (a + d))) \\ &= ((a' + d') + (b' + c')) + ((a + d) + x) \end{aligned}$$

and the right hand side:

$$\begin{aligned} & ((b' + c') + (b + a')) + (c + d') \\ &= (b' + c') + ((b + a') + (c + d')) \\ &= (b' + c') + ((a' + d') + (b + c)) \\ &= ((a' + d') + (b' + c')) + (b + c) \end{aligned}$$

So by cancellation  $(a + d) + x = b + c$  and  $a + d < b + c$ .  $\square$

**Lemma 1.21.**

$$\forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} ([\langle a, b \rangle] < [\langle c, d \rangle] \leftrightarrow a + d < b + c)$$

*Proof.* Let arbitrary  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$ . First suppose  $[\langle a, b \rangle] < [\langle c, d \rangle]$ . Then by definition there exists  $\langle a_0, b_0 \rangle, \langle c_0, d_0 \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $[\langle a, b \rangle] = [\langle a_0, b_0 \rangle]$ ,  $[\langle c, d \rangle] = [\langle c_0, d_0 \rangle]$  and  $a_0 + d_0 < b_0 + c_0$ . It then immediately follows by lemma 1.20 that  $a + d < b + c$ .

The reverse direction is straightforward. Suppose  $a + d < b + c$  and it immediately follows by definition that  $[\langle a, b \rangle] < [\langle c, d \rangle]$ .  $\square$

*Proof.* (range/domain bound and well-definedness) We start by showing  $<$  is a relation on  $\mathbb{Z}$ ,  $< \subseteq \mathbb{Z} \times \mathbb{Z}$ . Let arbitrary  $x \in <$ . Then by definition there exists  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \in \mathbb{Z} \times \mathbb{Z}$ .

Next well-definedness. That is,

$$\begin{aligned} & \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N} ( \\ & \quad ([\langle a, b \rangle] = [\langle a', b' \rangle] \wedge [\langle c, d \rangle] = [\langle c', d' \rangle]) \\ & \quad \rightarrow ([\langle a, b \rangle] < [\langle c, d \rangle] \leftrightarrow [\langle a', b' \rangle] < [\langle c', d' \rangle])) \end{aligned}$$

Letting arbitrary  $\langle a, b \rangle, \langle c, d \rangle, \langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$ . Suppose that  $[\langle a, b \rangle] = [\langle a', b' \rangle]$  and  $[\langle c, d \rangle] = [\langle c', d' \rangle]$ , which then means  $a + b' = b + a'$  and  $c + d' = d + c'$ . Suppose first that  $[\langle a, b \rangle] < [\langle c, d \rangle]$ , which by lemma 1.21 means  $a + d < b + c$ . It follows by lemma 1.20 that  $a' + d' < b' + c'$  and by lemma 1.21 that  $[\langle a', b' \rangle] < [\langle c', d' \rangle]$ . An identical strategy proves the reverse direction.  $\square$

(next page)

**Definition 1.22.** The binary relation  $\leq$  on  $\mathbb{Z}$  is defined such that

$$\begin{aligned} \forall x (x \in \leq \leftrightarrow \exists \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} (x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \\ \wedge ([\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle])))) \end{aligned}$$

( $\mathbb{Z} \times \mathbb{Z}$  is a bounding set. Uniqueness follows from extensionality.)

**Lemma 1.23.**

$$\begin{aligned} \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} ([\langle a, b \rangle] \leq [\langle c, d \rangle] \\ \leftrightarrow ([\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle])) \end{aligned}$$

*Proof.* Let arbitrary  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$ . First suppose  $[\langle a, b \rangle] \leq [\langle c, d \rangle]$ . Then by definition there exists  $\langle a_0, b_0 \rangle, \langle c_0, d_0 \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $[\langle a, b \rangle] = [\langle a_0, b_0 \rangle]$ ,  $[\langle c, d \rangle] = [\langle c_0, d_0 \rangle]$ , and  $[\langle a_0, b_0 \rangle] < [\langle c_0, d_0 \rangle]$  or  $[\langle a_0, b_0 \rangle] = [\langle c_0, d_0 \rangle]$ . In the first case suppose  $[\langle a_0, b_0 \rangle] < [\langle c_0, d_0 \rangle]$ . It then follows from the well-definedness of  $<$  that  $[\langle a, b \rangle] < [\langle c, d \rangle]$ . In the other case suppose  $[\langle a_0, b_0 \rangle] = [\langle c_0, d_0 \rangle]$ . It immediately follows that  $[\langle a, b \rangle] = [\langle a_0, b_0 \rangle] = [\langle c_0, d_0 \rangle] = [\langle c, d \rangle]$ .

Now suppose  $[\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle]$ . It immediately follows by definition that  $[\langle a, b \rangle] \leq [\langle c, d \rangle]$ .  $\square$

*Proof.* (domain/range bound and well-definedness) First we show  $\leq \subseteq \mathbb{Z} \times \mathbb{Z}$ . Let arbitrary  $x \in \leq$ . Then there exists  $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{N} \times \mathbb{N}$  such that  $x = \langle [\langle a, b \rangle], [\langle c, d \rangle] \rangle \in \mathbb{Z} \times \mathbb{Z}$ .

Next well-definedness. That is

$$\begin{aligned} \forall \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c, d \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b' \rangle \in \mathbb{N} \times \mathbb{N} \forall \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N} ( \\ ([\langle a, b \rangle] = [\langle a', b' \rangle] \wedge [\langle c, d \rangle] = [\langle c', d' \rangle]) \\ \rightarrow ([\langle a, b \rangle] \leq [\langle c, d \rangle] \leftrightarrow [\langle a', b' \rangle] \leq [\langle c', d' \rangle])) \end{aligned}$$

Letting arbitrary  $\langle a, b \rangle, \langle c, d \rangle, \langle a', b' \rangle, \langle c', d' \rangle \in \mathbb{N} \times \mathbb{N}$ . Suppose that  $[\langle a, b \rangle] = [\langle a', b' \rangle]$  and  $[\langle c, d \rangle] = [\langle c', d' \rangle]$ . First suppose  $[\langle a, b \rangle] \leq [\langle c, d \rangle]$ . It follows by lemma 1.23 that  $[\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle]$ . In the first case where  $[\langle a, b \rangle] < [\langle c, d \rangle]$  it follows from the well-definedness of  $<$  that  $[\langle a', b' \rangle] < [\langle c', d' \rangle]$  and again by lemma 1.23 that  $[\langle a', b' \rangle] \leq [\langle c', d' \rangle]$ . For the other case where  $[\langle a, b \rangle] = [\langle c, d \rangle]$  we immediately have  $[\langle a', b' \rangle] = [\langle a, b \rangle] = [\langle c, d \rangle] = [\langle c', d' \rangle]$  and by lemma 1.23  $[\langle a', b' \rangle] \leq [\langle c', d' \rangle]$ . The reverse implication follows by the same argument.  $\square$

### 1.2.4 Properties

**Theorem 1.24.** *Let  $x, y, z \in \mathbb{Z}$ .*

1.  $(x + y) + z = x + (y + z)$  (*Associative Law for Addition*)
2.  $x + y = y + x$  (*Commutative Law for Addition*)
3.  $x + \hat{0} = x$  (*Identity Law for Addition*)
4.  $x + (-x) = \hat{0}$  (*Inverses Law for Addition*)
5.  $(xy)z = x(yz)$  (*Associative Law for Multiplication*)
6.  $xy = yx$  (*Commutative Law for Multiplication*)
7.  $x \cdot \hat{1} = x$  (*Identity Law for Multiplication*)
8.  $x(y + z) = xy + xz$  (*Distributive Law*)
9. If  $xy = \hat{0}$ , then  $x = \hat{0}$  or  $y = \hat{0}$  (*No Zero Divisors Law*)
10. Precisely one of  $x < y$  or  $x = y$  or  $x > y$  holds (*Trichotomy Law*)
11. If  $x < y$  and  $y < z$ , then  $x < z$  (*Transitive Law*)
12. If  $x < y$  then  $x + z < y + z$  (*Addition Law for Order*)
13. If  $x < y$  and  $z > \hat{0}$ , then  $xz < yz$  (*Multiplication Law for Order*)
14.  $\hat{0} \neq \hat{1}$  (*Non-Triviality*)

*Proof.* (1) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} ((x + y) + z = x + (y + z))$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . Then there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . We have

$$\begin{aligned} (x + y) + z &= ([\langle a, b \rangle] + [\langle c, d \rangle]) + [\langle e, f \rangle] \\ &= [\langle a + c, b + d \rangle] + [\langle e, f \rangle] \\ &= [\langle (a + c) + e, (b + d) + f \rangle] \\ &= [\langle a + (c + e), b + (d + f) \rangle] \\ &= [\langle a, b \rangle] + [\langle c + e, d + f \rangle] \\ &= [\langle a, b \rangle] + ([\langle c, d \rangle] + [\langle e, f \rangle]) \\ &= x + (y + z) \end{aligned}$$

Where the fourth equality follows from associativity of  $\mathbb{N}$ . □

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*Proof.* (2) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (x + y = y + x)$$

Let arbitrary  $x, y \in \mathbb{Z}$ . Then there exists  $a, b, c, d \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$  and  $y = [\langle c, d \rangle]$ . It follows that

$$\begin{aligned} x + y &= [\langle a, b \rangle] + [\langle c, d \rangle] = [\langle a + c, b + d \rangle] \\ &= [\langle c + a, d + b \rangle] = [\langle c, d \rangle] + [\langle a, b \rangle] = y + x \quad \square \end{aligned}$$

*Proof.* (3) We want to show

$$\forall x \in \mathbb{Z} (x + \hat{0} = x)$$

Let arbitrary  $x \in \mathbb{Z}$ . Then there exists  $a, b \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ . We have  $a + (b + 1) = b + (a + 1)$  and therefore  $\langle a, b \rangle \sim \langle a + 1, b + 1 \rangle$ . So

$$x = [\langle a, b \rangle] = [\langle a + 1, b + 1 \rangle] = [\langle a, b \rangle] + [\langle 1, 1 \rangle] = x + \hat{0} \quad \square$$

*Proof.* (4) We want to show

$$\forall x \in \mathbb{Z} (x + (-x) = \hat{0})$$

Let arbitrary  $x \in \mathbb{Z}$ . Then there exists  $a, b \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ . Then by definition  $-x = [\langle b, a \rangle]$ . We have  $x + (-x) = [\langle a, b \rangle] + [\langle b, a \rangle] = [\langle a + b, b + a \rangle]$ . We also know  $(a + b) + 1 = (b + a) + 1$ , meaning  $\langle a + b, b + a \rangle \sim \langle 1, 1 \rangle$ . Putting it all together we have

$$x + (-x) = [\langle a + b, b + a \rangle] = [\langle 1, 1 \rangle] = \hat{0} \quad \square$$

*Proof.* (5) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} ((xy)z = x(yz))$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . Then there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . We have

$$\begin{aligned} (xy)z &= [\langle ac + bd, ad + bc \rangle] \cdot [\langle e, f \rangle] \\ &= [\langle (ac + bd)e + (ad + bc)f, (ac + bd)f + (ad + bc)e \rangle] \\ &= [\langle ((ac)e + (bd)e) + ((ad)f + (bc)f), ((ac)f + (bd)f) + ((ad)e + (bc)e) \rangle] \\ &= [\langle (a(ce) + b(df)) + (b(de) + a(cf)), (a(cf) + b(de)) + (b(ce) + a(df)) \rangle] \\ &= [\langle a(ce + df) + b(cf + de), a(cf + de) + b(ce + df) \rangle] \\ &= [\langle a, b \rangle] \cdot [\langle ce + df, cf + de \rangle] \\ &= [\langle a, b \rangle] \cdot ([\langle c, d \rangle] \cdot [\langle e, f \rangle]) \\ &= x(yz) \quad \square \end{aligned}$$

(next page)



*Proof.* (6) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (xy = yx)$$

Let arbitrary  $x, y \in \mathbb{Z}$ . Then there exists  $a, b, c, d \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$  and  $y = [\langle c, d \rangle]$ . It follows that

$$xy = [\langle ac + bd, ad + bc \rangle] = [\langle ca + db, cb + da \rangle] = [\langle c, d \rangle] \cdot [\langle a, b \rangle] = yx \quad \square$$

*Proof.* (7) We want to show

$$\forall x \in \mathbb{Z} (x \cdot \hat{1} = x)$$

Let arbitrary  $x \in \mathbb{Z}$ . Then there exists  $a, b \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ . We have

$$\begin{aligned} x \cdot \hat{1} &= [\langle a, b \rangle] \cdot [\langle 1 + 1, 1 \rangle] = [\langle a(1 + 1) + b \cdot 1, a \cdot 1 + b(1 + 1) \rangle] \\ &= [\langle (a + a) + b, a + (b + b) \rangle] \end{aligned}$$

We also have  $a + (a + (b + b)) = b + ((a + a) + b)$ , meaning  $\langle a, b \rangle \sim \langle (a + a) + b, a + (b + b) \rangle$ . Combining these two facts we have

$$x \cdot \hat{1} = [\langle (a + a) + b, a + (b + b) \rangle] = [\langle a, b \rangle] = x \quad \square$$

*Proof.* (8) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x(y + z) = xy + xz)$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . Then there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . We have

$$\begin{aligned} x(y + z) &= [\langle a, b \rangle] \cdot [\langle c + e, d + f \rangle] \\ &= [\langle a(c + e) + b(d + f), a(d + f) + b(c + e) \rangle] \\ &= [\langle (ac + bd) + (ae + bf), (ad + bc) + (af + be) \rangle] \\ &= [\langle ac + bd, ad + bc \rangle] + [\langle ae + bf, af + be \rangle] \\ &= [\langle a, b \rangle] \cdot [\langle c, d \rangle] + [\langle a, b \rangle] \cdot [\langle e, f \rangle] \\ &= xy + xz \quad \square \end{aligned}$$

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**Lemma 1.25.**  $\forall a \in \mathbb{N} \forall b \in \mathbb{N} ([\langle a, b \rangle] = \hat{0} \leftrightarrow a = b)$

*Proof.* Let arbitrary  $a, b \in \mathbb{N}$ . First suppose that  $[\langle a, b \rangle] = \hat{0} = [\langle 1, 1 \rangle]$ . Then  $a + 1 = b + 1$  and by cancellation  $a = b$ . Next suppose  $a = b$ . Then  $a + 1 = b + 1$  and  $\langle a, b \rangle \sim \langle 1, 1 \rangle$ . We then have  $[\langle a, b \rangle] = [\langle 1, 1 \rangle] = \hat{0}$ .  $\square$

*Proof.* (9) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} (xy = \hat{0} \rightarrow (x = \hat{0} \vee y = \hat{0}))$$

Let arbitrary  $x, y \in \mathbb{Z}$  and suppose that  $xy = \hat{0}$ . The statement after the implication is logically equivalent to  $x \neq \hat{0} \rightarrow y = \hat{0}$ . Suppose  $x \neq \hat{0}$ . There exists  $a, b, c, d \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$  and  $y = [\langle c, d \rangle]$ . We then have  $xy = [\langle ac + bd, ad + bc \rangle] = \hat{0}$ ; by the previous lemma  $ac + bd = ad + bc$ . That lemma also implies  $a \neq b$ ; so by trichotomy of the natural numbers we have  $a < b$  or  $b < a$ .

In the first case suppose  $a < b$ . Then there exists  $g \in \mathbb{N}$  such that  $a + g = b$ . We then have

$$(ac + ad) + gd = ac + (a + g)d = ad + (a + g)c = (ac + ad) + gc$$

Cancellation of addition on the naturals allows us to conclude that  $gd = gc$ . Cancellation of multiplication allows us to conclude  $c = d$ . By the previous lemma this implies  $y = [\langle c, d \rangle] = \hat{0}$ .

In the other case suppose  $b < a$ . Then there exists  $g \in \mathbb{N}$  such that  $b + g = a$ . We then have

$$(bc + bd) + gc = (b + g)c + bd = (b + g)d + bc = (bc + bd) + gd$$

Similarly again we conclude  $gc = gd$  and  $c = d$ , which then implies  $y = [\langle c, d \rangle] = \hat{0}$ .  $\square$

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*Proof.* (10) We want to show

$$\begin{aligned} \forall x \in \mathbb{Z} \forall y \in \mathbb{Z} ((x < y \vee x = y \vee y < x) \wedge \\ \neg(x < y \wedge x = y) \wedge \neg(x < y \wedge y < x) \wedge \neg(x = y \wedge y < x)) \end{aligned}$$

Let arbitrary  $x, y \in \mathbb{Z}$ . Then there exists  $a, b, c, d \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$  and  $y = [\langle c, d \rangle]$ . We start with the latter three statements. First suppose  $x < y$  and  $x = y$ . By lemma 1.21  $a + d < b + c$  and by definition  $a + d = b + c$ ; this contradicts trichotomy for the naturals. Next suppose  $x < y$  and  $y < x$ . Again this then implies  $a + d < b + c$  and  $b + c < a + d$  which contradicts trichotomy. Supposing  $x = y$  and  $y < x$  also contradicts trichotomy in a similar manner.

Finally the former statement. We know that  $a + d, b + c \in \mathbb{N}$ . By trichotomy we must have

$$a + d < b + c \vee a + d = b + c \vee b + c < a + d$$

which by lemma 1.21 means

$$[\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle] \vee [\langle c, d \rangle] < [\langle a, b \rangle]$$

which is equivalent to our desired result.  $\square$

*Proof.* (11) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x < y \wedge y < z \rightarrow x < z)$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . Suppose  $x < y$  and  $y < z$ . Then by definition there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . By lemma 1.21 we can assert that  $a + d < b + c$  and  $c + f < d + e$ . By definition there then exists  $g, g' \in \mathbb{N}$  such that  $(a + d) + g = b + c$  and  $(c + f) + g' = d + e$ . The former, along with cancellation, associativity and commutativity, allows us to assert

$$((a + d) + g) + f = ((a + d) + f) + g = (b + c) + f$$

So  $(a + d) + f < (b + c) + f$ . Similarly the latter allows us to assert

$$((c + f) + g') + b = ((b + c) + f) + g' = (d + e) + b = b + (d + e)$$

So  $(b + c) + f < b + (d + e)$ . By transitivity of the naturals we then have

$$(a + d) + f < b + (d + e)$$

Then there exists  $g_0 \in \mathbb{N}$  such that  $((a + d) + f) + g_0 = b + (d + e)$ . So we have

$$d + ((a + f) + g_0) = ((a + d) + f) + g_0 = b + (d + e) = d + (b + e)$$

which by cancellation means

$$(a + f) + g_0 = b + e$$

and  $a + f < b + e$ . By lemma 1.21 this means  $[\langle a, b \rangle] < [\langle e, f \rangle]$  and  $x < z$ .  $\square$

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*Proof.* (12) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x < y \rightarrow x + z < y + z)$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . By definition there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . Suppose  $x < y$ . By lemma 1.21 we then have  $a + d < b + c$ . By theorem 1.10 part 4 and commutativity and associativity of the naturals, we can assert

$$(a + e) + (d + f) = (a + d) + (e + f) < (b + c) + (e + f) = (b + f) + (c + e)$$

which by lemma 1.21 and the definition of addition, implies

$$[\langle a, b \rangle] + [\langle e, f \rangle] = [\langle a + e, b + f \rangle] < [\langle c + e, d + f \rangle] = [\langle c, d \rangle] + [\langle e, f \rangle]$$

which is equivalent to our desired result.  $\square$

*Proof.* (13) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x < y \wedge z > \hat{0} \rightarrow xz < yz)$$

Let arbitrary  $x, y, z \in \mathbb{Z}$ . By definition there exists  $a, b, c, d, e, f \in \mathbb{N}$  such that  $x = [\langle a, b \rangle]$ ,  $y = [\langle c, d \rangle]$ , and  $z = [\langle e, f \rangle]$ . Suppose that  $x < y$  and  $z > \hat{0}$ . By lemma 1.21 we have  $a + d < b + c$  and  $1 + f < 1 + e$ . By theorem 1.10 part 4 we have  $f < e$ , so there exists  $k \in \mathbb{N}$  such that  $f + k = e$ . By theorem 1.10 part 6 we have  $k(a + d) < k(b + c)$ . Part 4 of the same theorem allows us to then assert

$$k(a + d) + ((a + d)f + (b + c)f) < k(b + c) + ((a + d)f + (b + c)f)$$

The right hand side is equivalent to

$$\begin{aligned} & k(a + d) + ((a + d)f + (b + c)f) \\ &= (ka + kd) + ((af + df) + (bf + cf)) \\ &= ((ka + kd) + (af + df)) + (bf + cf) \\ &= (a(k + f) + d(k + f)) + (bf + cf) \\ &= (ae + de) + (bf + cf) \\ &= (ae + bf) + (cf + de) \end{aligned}$$

Similarly the left hand side is equivalent to

$$\begin{aligned} & k(b + c) + ((a + d)f + (b + c)f) \\ &= (kb + kc) + ((af + df) + (bf + cf)) \\ &= (b(k + f) + c(k + f)) + (af + df) \\ &= (be + ce) + (af + df) \\ &= (af + be) + (ce + df) \end{aligned}$$

(next page)

We can then conclude that  $(ae + bf) + (cf + de) < (af + be) + (ce + df)$ . By lemma 1.21 and the definition of multiplication on the integers we then have

$$[\langle a, b \rangle] \cdot [\langle e, f \rangle] = [\langle ae + bf, af + be \rangle] < [\langle ce + df, cf + de \rangle] = [\langle c, d \rangle] \cdot [\langle e, f \rangle]$$

Which is equivalent to our desired conclusion.  $\square$

*Proof.* (14) We want to show  $\hat{0} \neq \hat{1}$ . Suppose for contradiction that they are equal, meaning  $[\langle 1, 1 \rangle] = [\langle 1+1, 1 \rangle]$ . By definition we then have  $1+1 = 1+(\hat{1})$ , which by cancellation implies  $1 = 1 + 1$ , contradicting part 5 of theorem 1.7.  $\square$

### 1.2.5 Joining the naturals and the integers

**Definition 1.26.** Let  $x \in \mathbb{Z}$ .  $x$  is **positive** if  $x > \hat{0}$ .  $x$  is **negative** if  $x < \hat{0}$ .

**Theorem 1.27.** Let  $i : \mathbb{N} \rightarrow \mathbb{Z}$  be defined by  $i(n) = [\langle n + 1, 1 \rangle]$  for all  $n \in \mathbb{N}$ .

1. The function  $i : \mathbb{N} \rightarrow \mathbb{Z}$  is injective.

2.  $i(\mathbb{N}) = \{x \in \mathbb{Z} \mid x > \hat{0}\}$

3.  $i(1) = \hat{1}$

4. Let  $a, b \in \mathbb{N}$ . Then

$$(a) \ i(a + b) = i(a) + i(b)$$

$$(b) \ i(ab) = i(a)i(b)$$

$$(c) \ a < b \text{ if and only if } i(a) < i(b)$$

*Proof.* (1) Defining  $i$  as the set such that

$$\forall x (x \in i \leftrightarrow \exists a \in \mathbb{N} \exists b (x = \langle a, b \rangle \wedge b = [\langle a + 1, 1 \rangle]))$$

Existence of such a set can be shown using the subset axiom, with  $\mathbb{N} \times \mathbb{Z}$  as a bounding set. We start by showing functionality. Let arbitrary  $a \in \text{dom}(i)$ . By definition there exists  $b$  such that  $\langle a, b \rangle \in i$ ; as defined this implies  $a \in \mathbb{N}$  and  $b = [\langle a + 1, 1 \rangle]$ . Let  $b'$  be arbitrary such that  $\langle a, b' \rangle \in i$ . Again by definition we immediately have  $b' = [\langle a + 1, 1 \rangle] = b$ .

Next the domain bound  $\text{dom}(i) = \mathbb{N}$ . Let  $a$  be arbitrary and suppose that  $a \in \text{dom}(i)$ . By definition there then exists  $b$  such that  $\langle a, b \rangle \in i$  and it immediately follows that  $a \in \mathbb{N}$ . Now suppose  $a \in \mathbb{N}$ . By definition  $[\langle a + 1, 1 \rangle] \in \mathbb{Z}$  and  $\langle a, [\langle a + 1, 1 \rangle] \rangle \in i$ . So  $a \in \text{dom}(i)$ .

Next the range bound  $\text{ran}(i) \subseteq \mathbb{Z}$ . Let arbitrary  $b \in \text{ran}(i)$ . Then by definition there exists  $a$  such that  $\langle a, b \rangle \in i$ . It immediately follows that  $a \in \mathbb{N}$  and  $b = [\langle a + 1, 1 \rangle] \in \mathbb{Z}$ .

Finally we show injectivity. Let arbitrary  $b \in \text{ran}(i)$ . By definition there exists  $a$  such that  $\langle a, b \rangle \in i$ ; it also follows that  $a \in \mathbb{N}$  and  $b = [\langle a + 1, 1 \rangle]$ . Let  $a'$  be arbitrary and suppose  $\langle a', b \rangle \in i$ ; similarly we have  $a' \in \mathbb{N}$  and  $b = [\langle a' + 1, 1 \rangle]$ . So  $[\langle a + 1, 1 \rangle] = [\langle a' + 1, 1 \rangle]$ , which implies  $\langle a + 1, 1 \rangle \sim \langle a' + 1, 1 \rangle$  and  $(a + 1) + 1 = 1 + (a' + 1)$ . By commutativity and cancellation we then have  $a = a'$ .  $\square$

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*Proof.* (2) We show  $\{i(n) \mid n \in \mathbb{N}\} = \{x \in \mathbb{Z} \mid x > \hat{0}\}$ . More specifically, denoting the former as  $A$  and the latter  $B$ , they are the sets such that

$$\begin{aligned}\forall x(x \in A &\leftrightarrow \exists n \in \mathbb{N}(x = i(n))) \\ \forall x(x \in B &\leftrightarrow x \in \mathbb{Z} \wedge x > \hat{0})\end{aligned}$$

Existence of  $A$  can be shown using the subset axiom with  $\mathbb{Z}$  as a bounding set. Existence of  $B$  is immediately assured by the subset axiom. We want to show  $A = B$ .

Let  $x$  be arbitrary and suppose  $x \in A$ . Then there exists  $n \in \mathbb{N}$  such that  $x = i(n)$ . So  $\langle n, x \rangle \in i$  and  $x \in \text{ran}(i) \subseteq \mathbb{Z}$ . By definition  $x = [\langle n+1, 1 \rangle]$ . Since we know  $(1+1) + n = 1 + (n+1)$  it follows that  $1+1 < 1 + (n+1)$  and by lemma 1.21 that  $\hat{0} = [\langle 1, 1 \rangle] < [\langle n+1, 1 \rangle] = x$ . So  $x \in B$ .

Now suppose  $x \in B$ . Then  $x \in \mathbb{Z}$  and  $x > \hat{0}$ . By definition there exists  $a, b \in \mathbb{N}$  such that  $x = [\langle a, b \rangle] > \hat{0}$ . By lemma 1.21  $1+b < 1+a$ , so there exists  $k \in \mathbb{N}$  such that  $(1+b) + k = 1+a$ ; associativity and cancellation allows us to conclude that  $b+k = a$  it is apparent that  $[\langle k+1, 1 \rangle] \in \mathbb{Z}$  and that by definition  $\langle k, [\langle k+1, 1 \rangle] \rangle \in i$ . We know  $(1+b) + k = 1+a$ ; so  $(k+1) + b = 1+a$ ,  $\langle k+1, 1 \rangle \sim \langle a, b \rangle$ , and  $x = [\langle a, b \rangle] = [\langle k+1, 1 \rangle] = i(k)$ , meaning  $x \in A$ .  $\square$

*Proof.* (3) We defined  $i$  such that

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{N} \exists b(x = \langle a, b \rangle \wedge b = [\langle a+1, 1 \rangle]))$$

By definition we have  $\langle 1, [\langle 1+1, 1 \rangle] \rangle \in i$ . So  $i(1) = [\langle 1+1, 1 \rangle] = \hat{1}$ .  $\square$

*Proof.* (4a)

$\square$